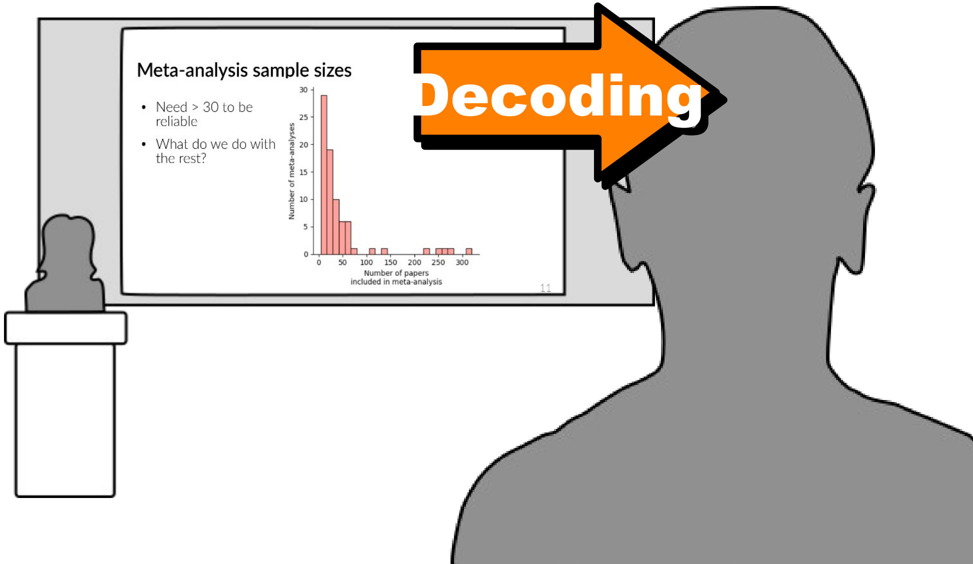
Introduction to Data Visualization

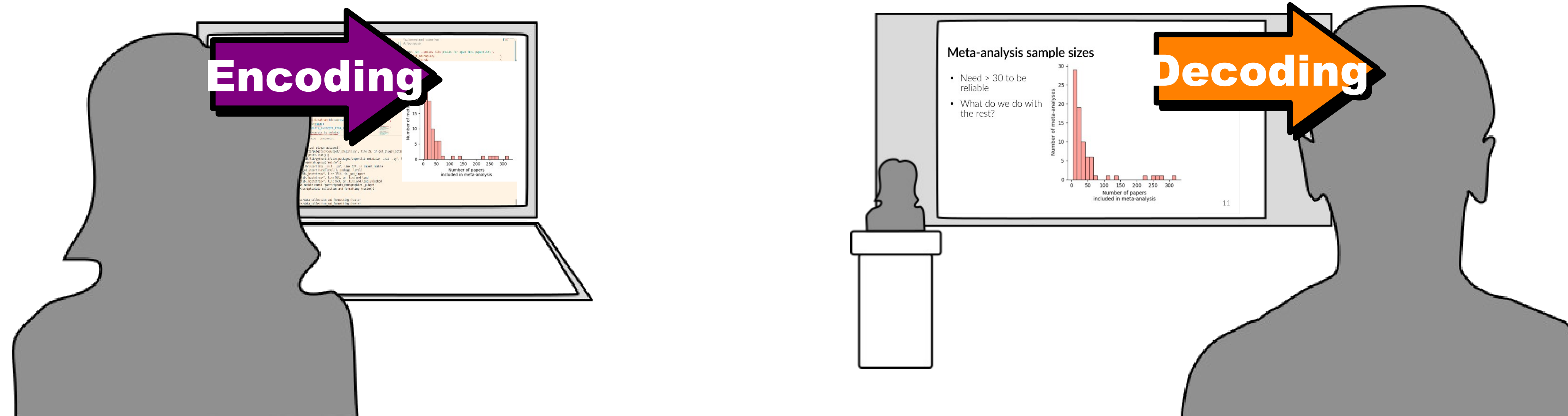
Part 1: Decoding

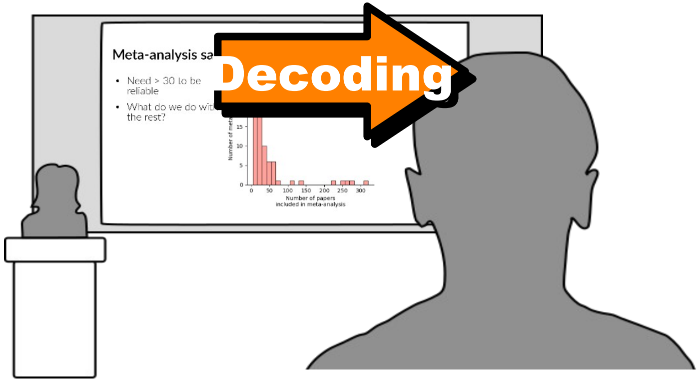
Johanna Bayer
Slides by Kendra Oudyk



Goal

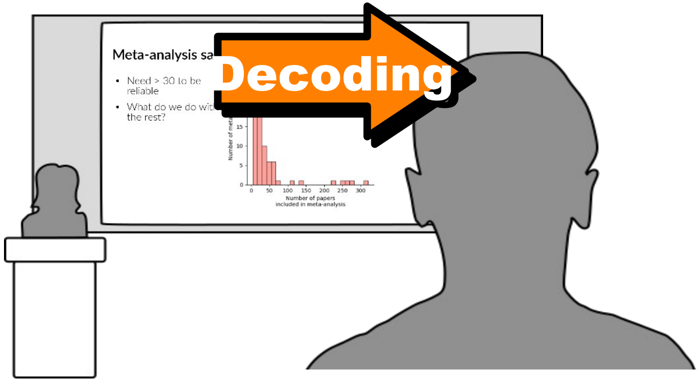
Use principles of visual **encoding** and **decoding**
to **efficiently** create visualizations
that are **effective** and **reproducible**





To plan an **effective visualization**, we need to think about

- **Message**
  - **What** we want to communicate
- **Perception**
  - **How best** to communicate it
- **Conventions**
  - **How** it's **usually** communicated



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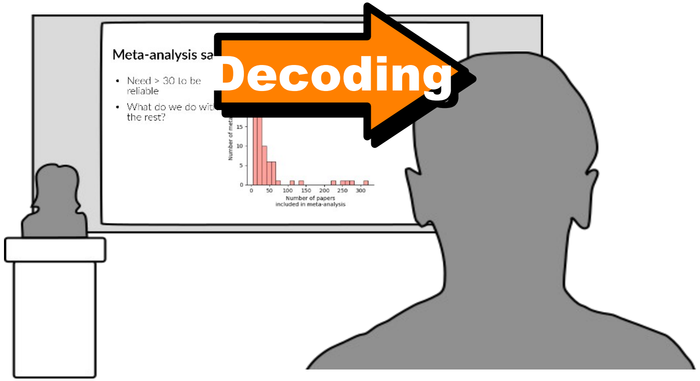
Message

- Raw data has no message
- Abstract it
  - "There are more males than females in science"
    --> A difference between magnitudes
  - "These brain areas activate together"
    --> A grouping / pattern



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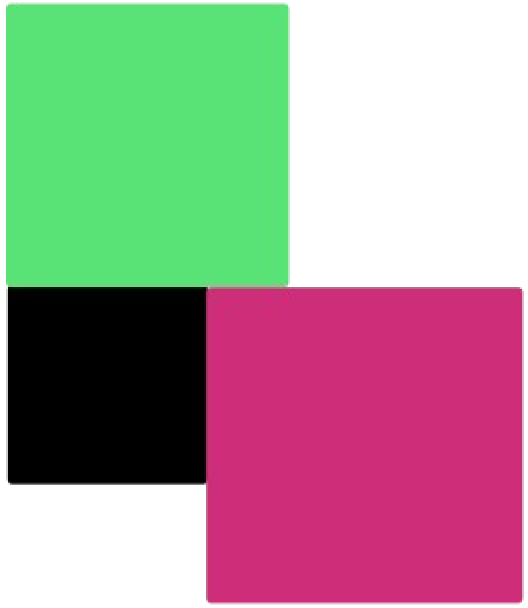
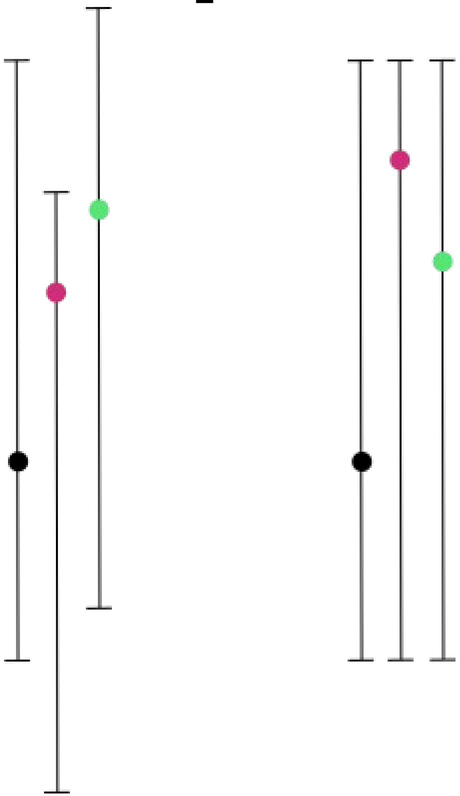
Which blue bar is bigger?

Which is 2x black, green or pink?





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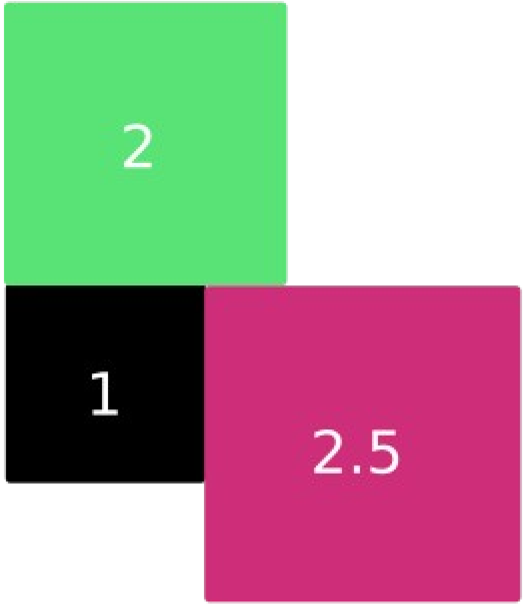
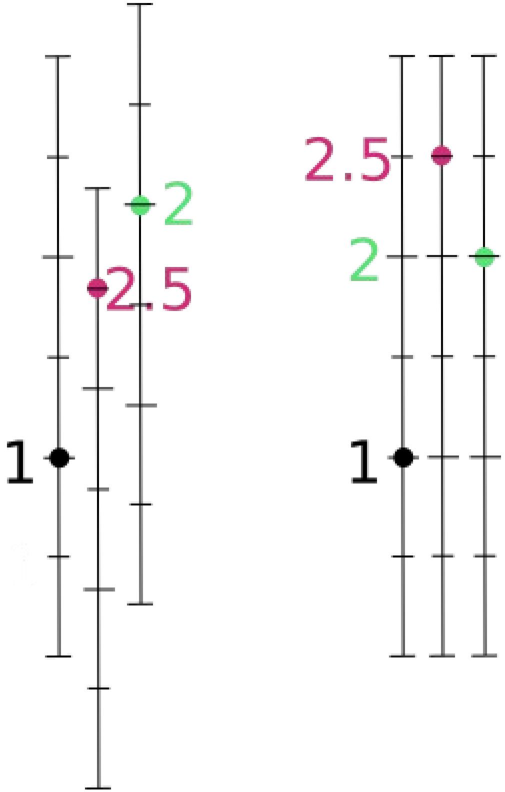
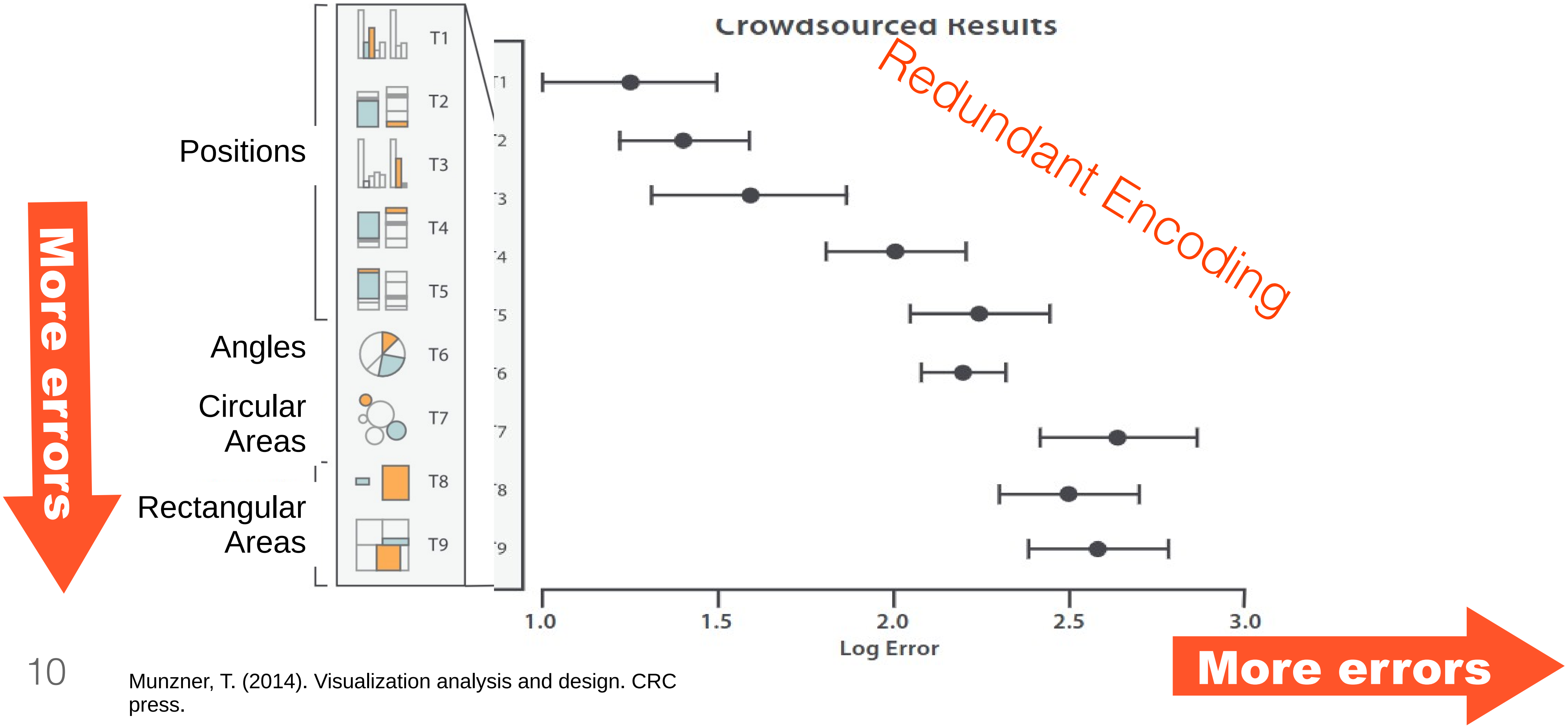




We’re better at judging aligned positions



Munzner, T. (2014). Visualization analysis and design. CRC press.

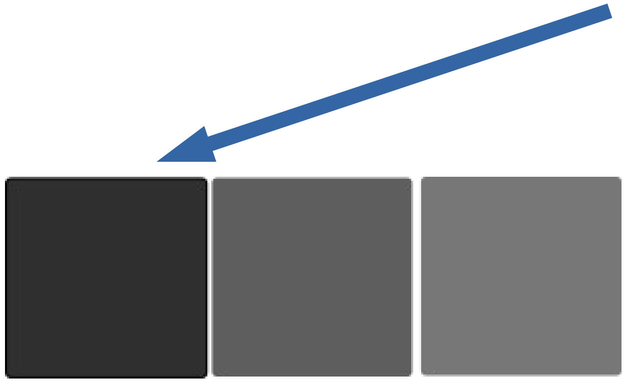
Decoding values vs. patterns

- “There are more males than females in science”

  --> A difference between **magnitudes**
- “These brain areas activate together”

  --> A grouping / **pattern**

Which is 2x lighter than the leftmost box?



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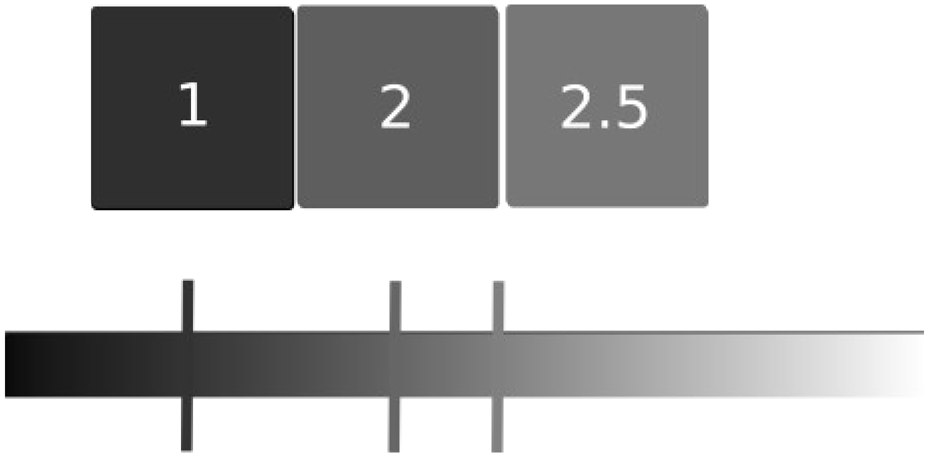
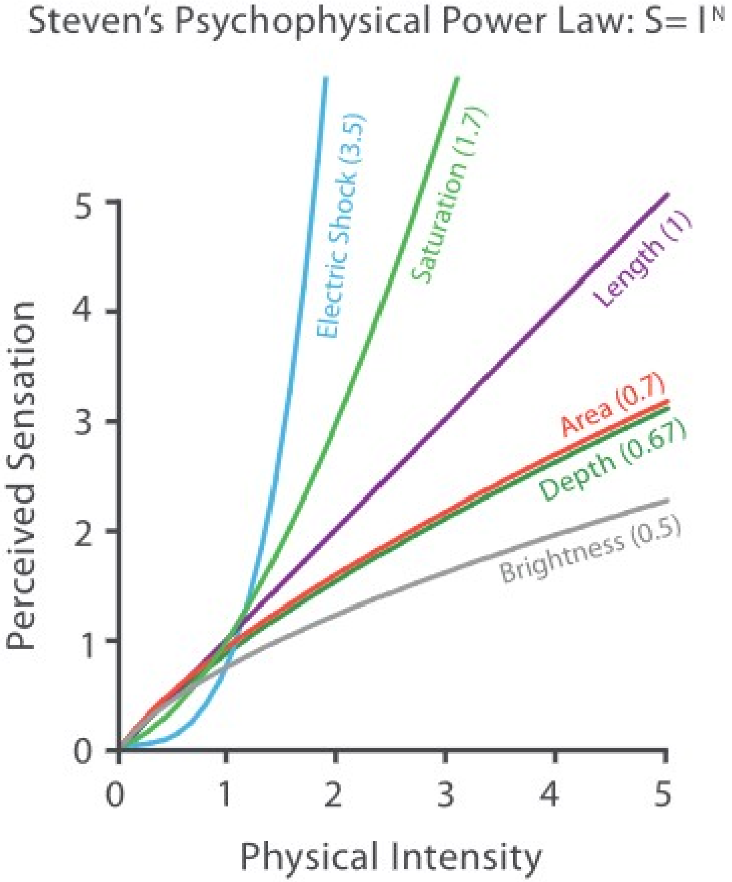


Physical intensity vs perceived intensity



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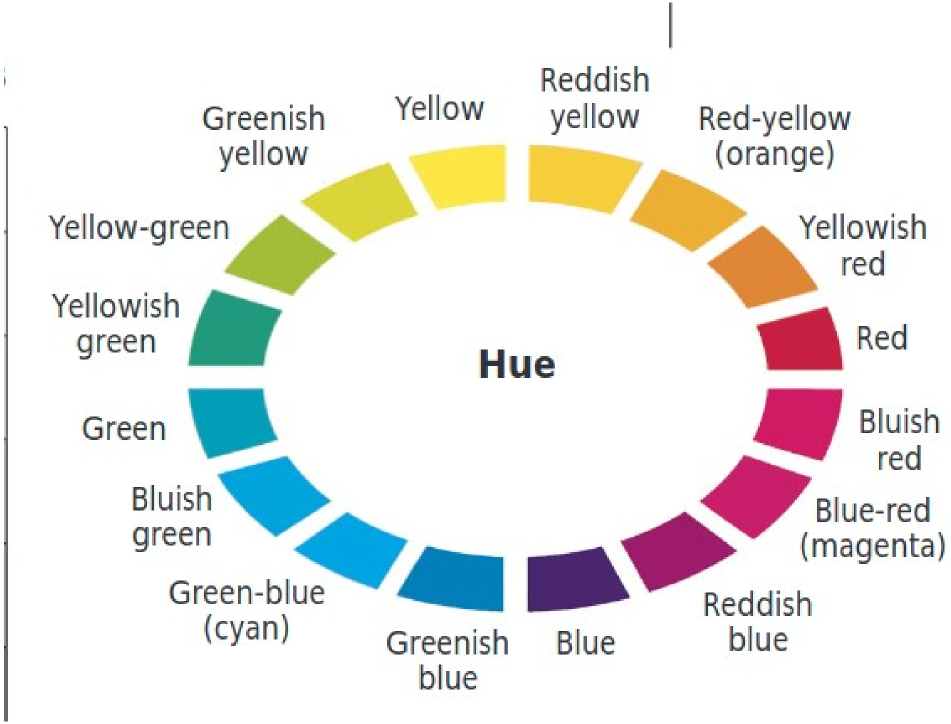
Color: salient but complicated

The HSL color model

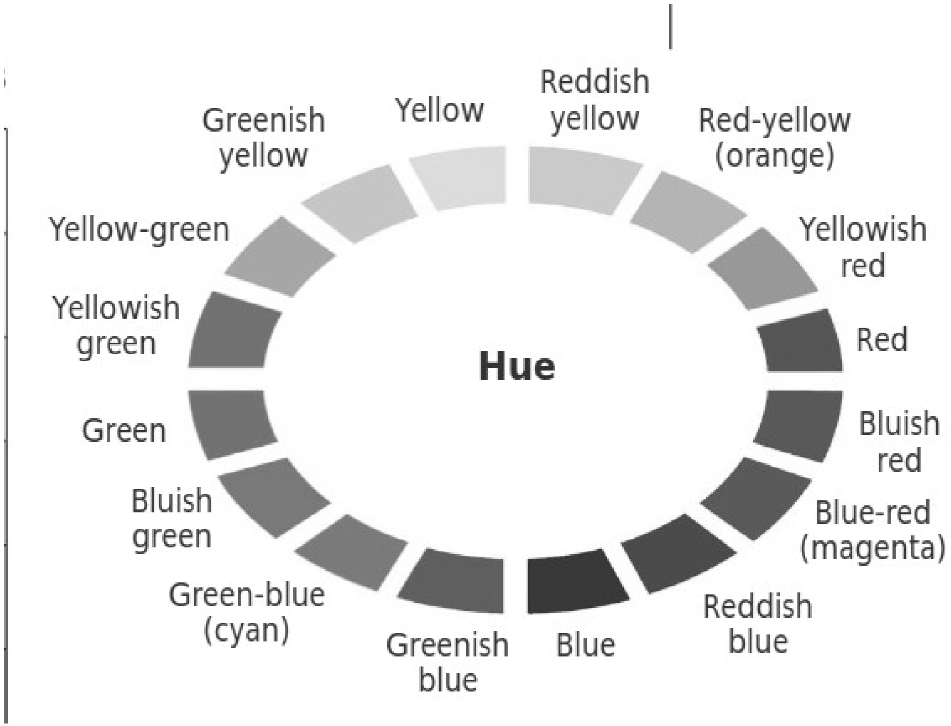
- H - Hue: the "type" of color, represented as an angle from 0–360°. Red at 0°, green at 120°, blue at 240°, and back to red at 360°
- S - Saturation: 0% is grey, 100% is the fully vivid version of that hue
- L – Lightness/Luminance: 0% is black, 50% is the "pure" color, 100% is white

Hue: how we talk about color



https://www.engr.colostate.edu/ECE666/Handouts/WritingPapers/UsingColorEffectively.pdf

Luminance: an important subconscious cue



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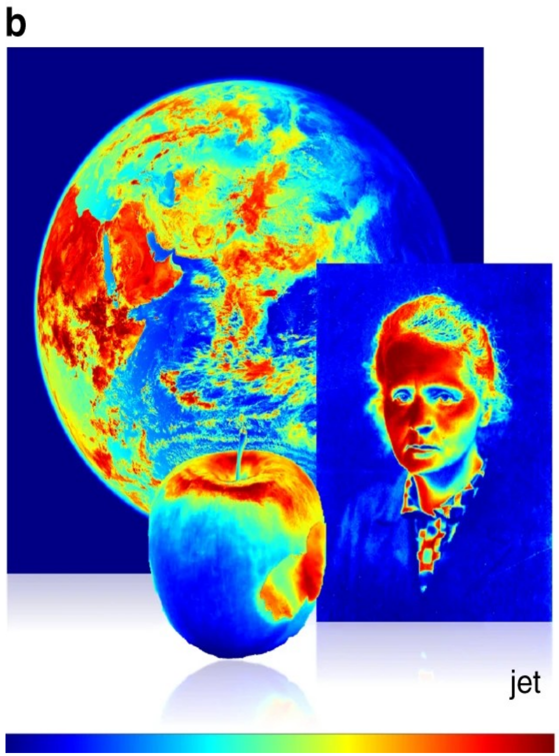
Named colors don't work well for ordered values



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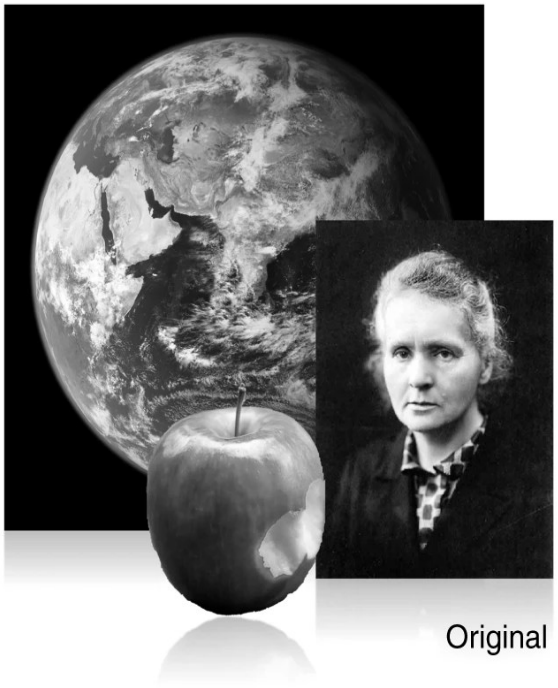

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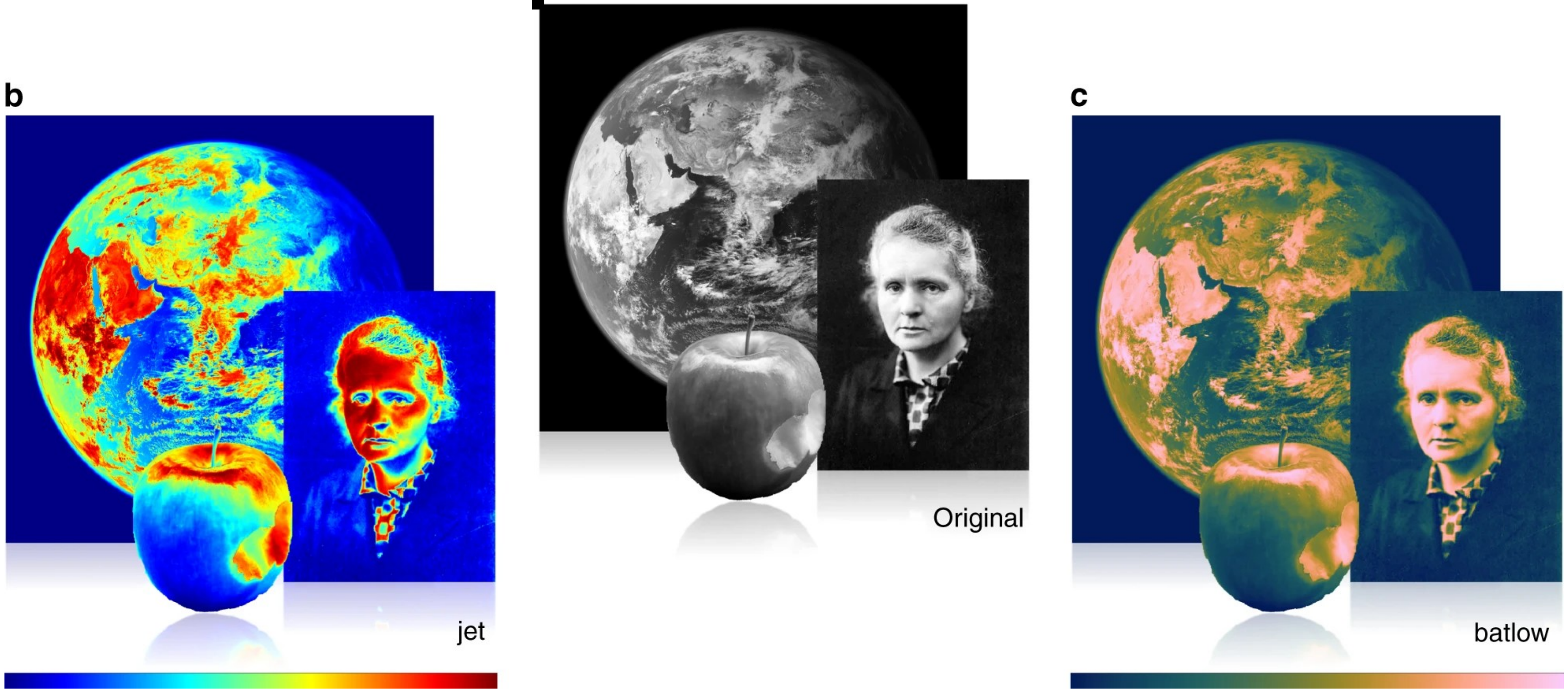
Certain colormaps do work



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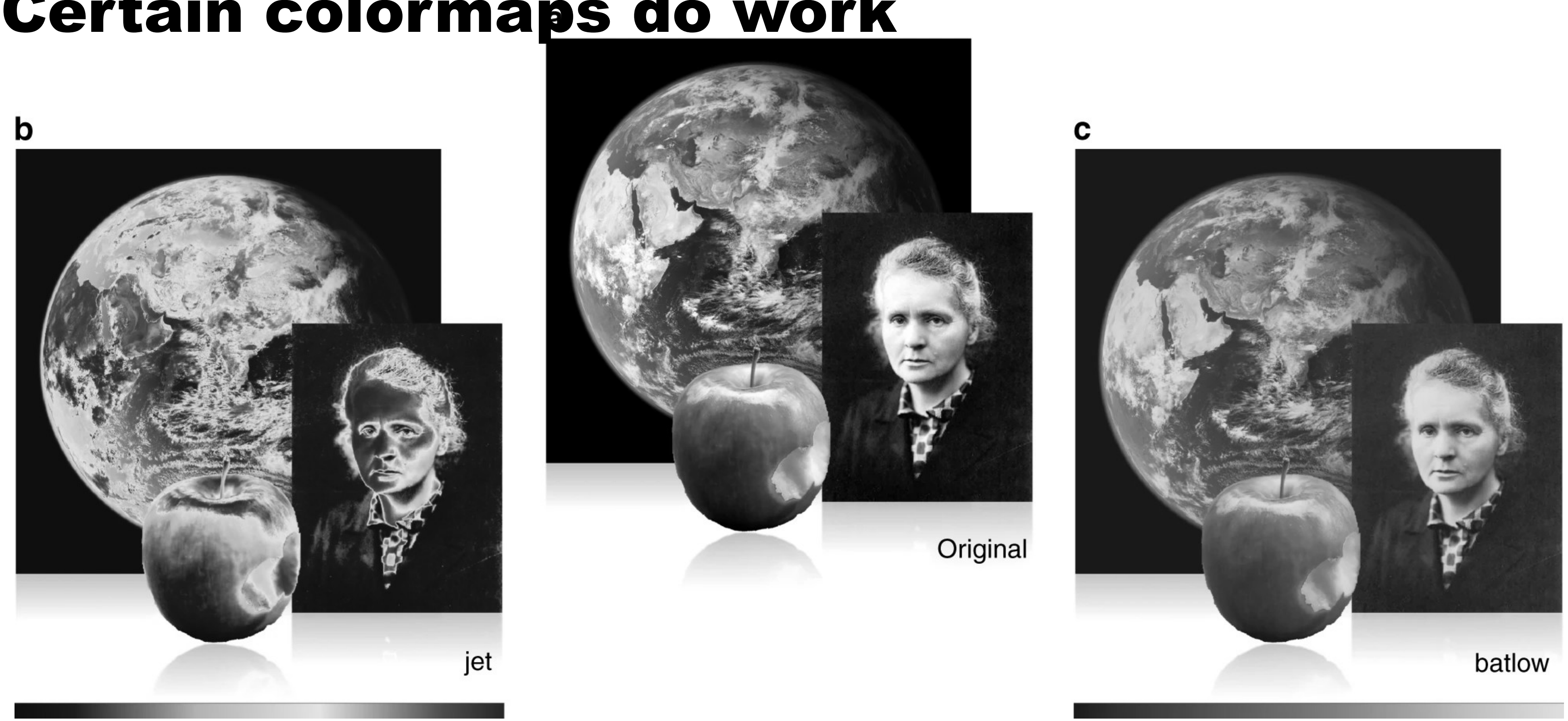

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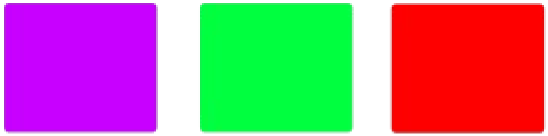
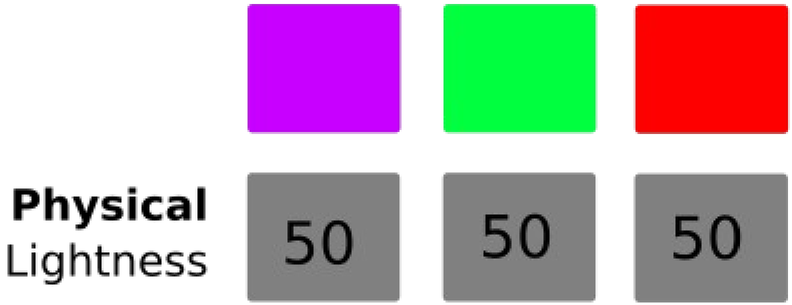
The brain’s sensitivity to luminance

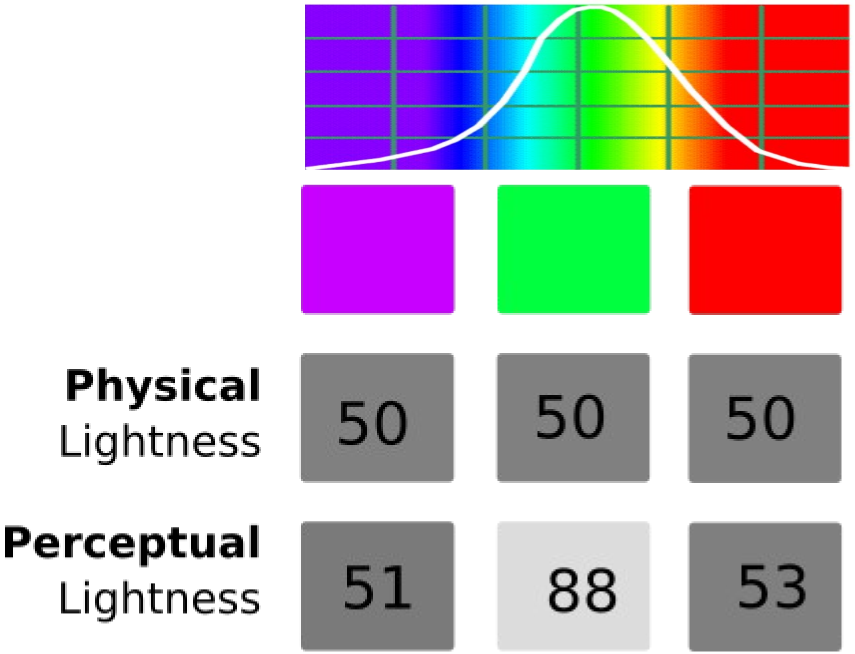
Which is lightest?




Physical
Lightness
50
50
50


Physical
Lightness
50
50
50
Perceptual
Lightness
51
88
53

Spectral sensitivity to luminance (lightness)



(upper figure) https://www.yorku.ca/eye/photopik.htm

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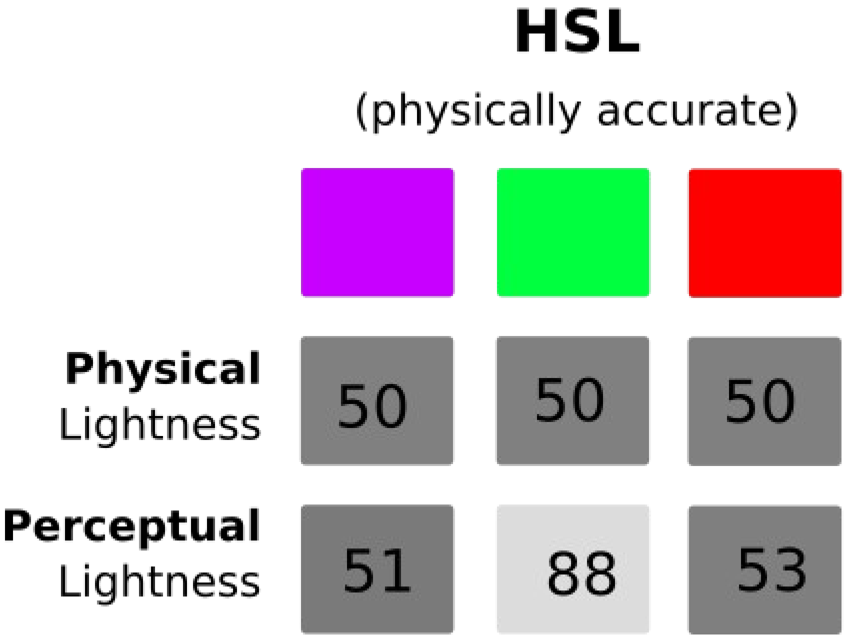

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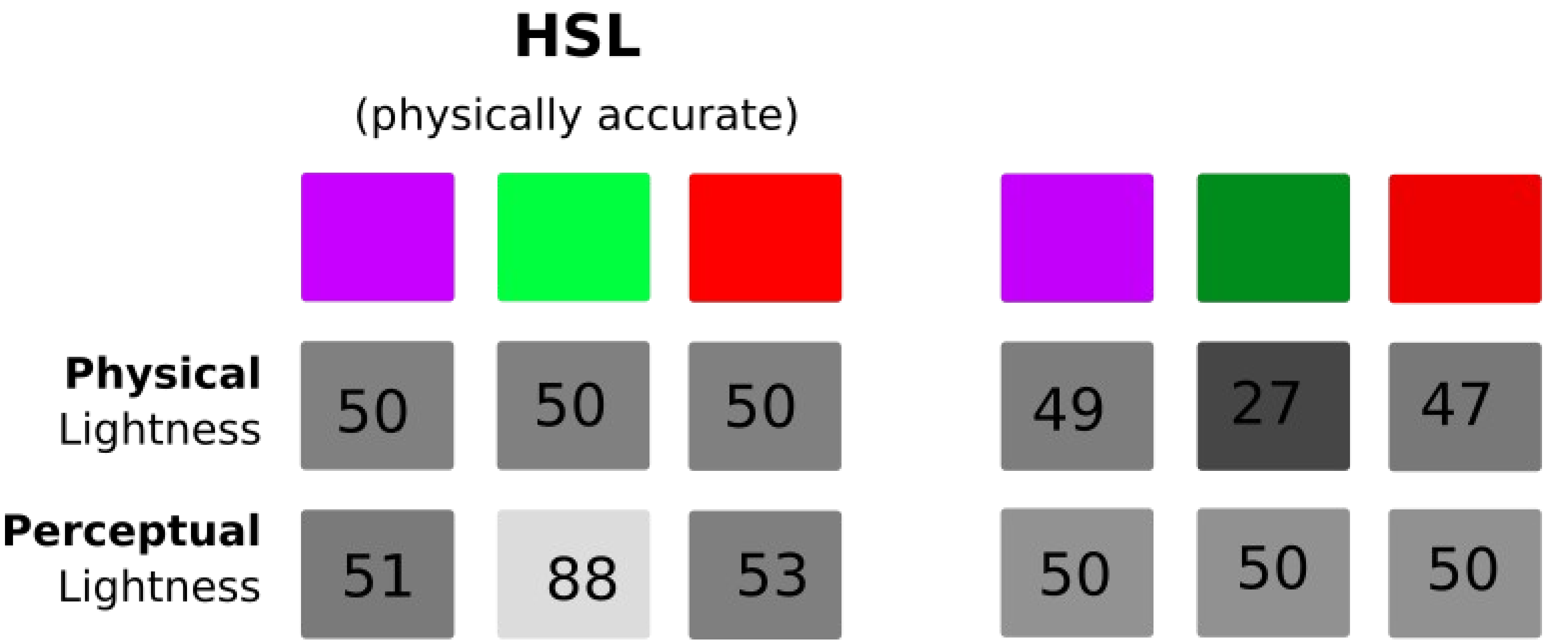

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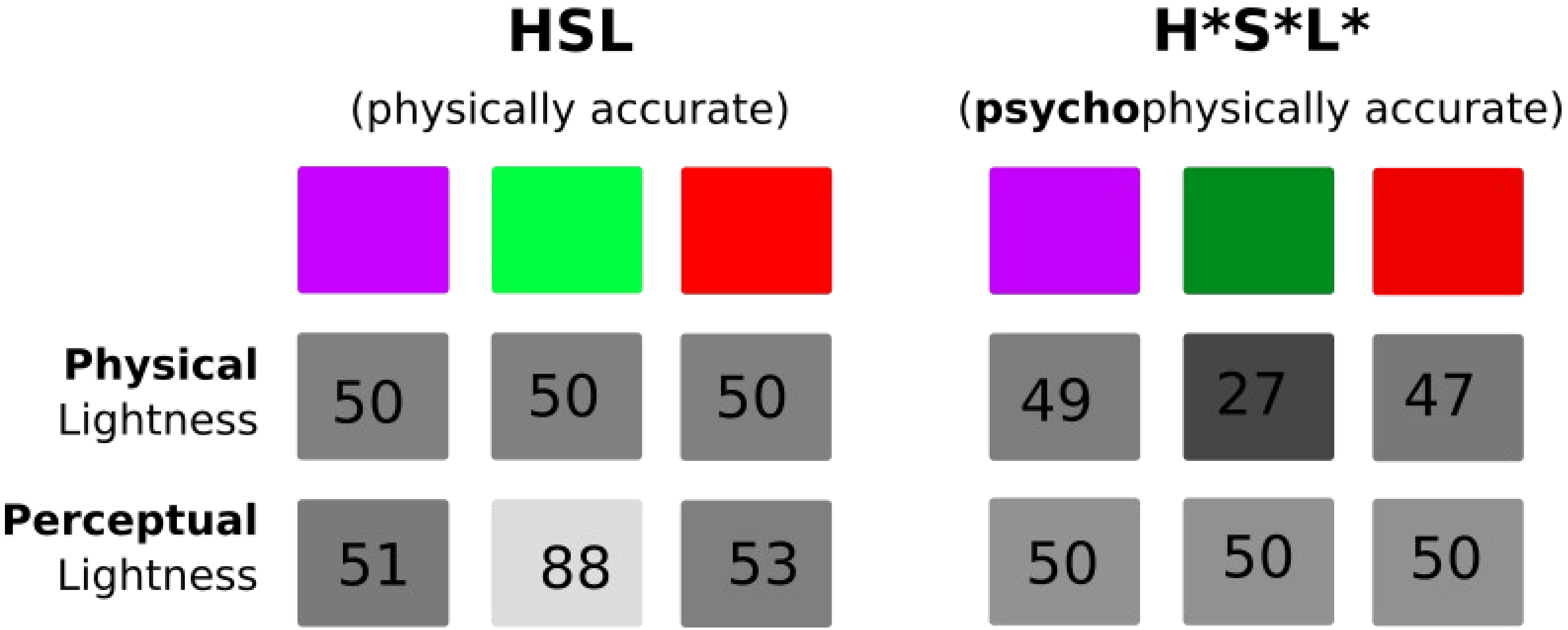
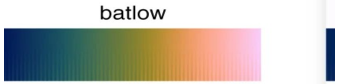
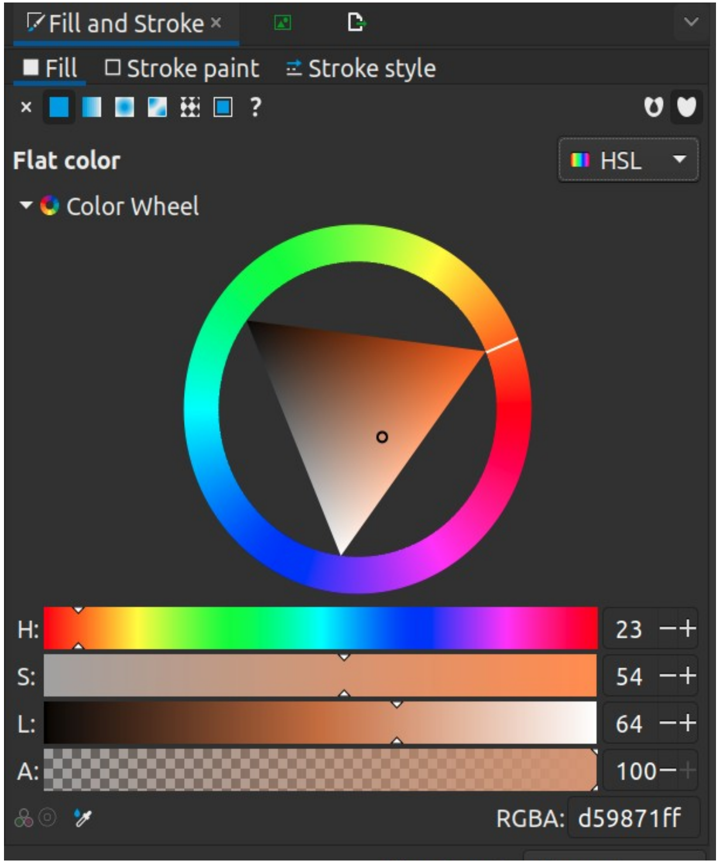
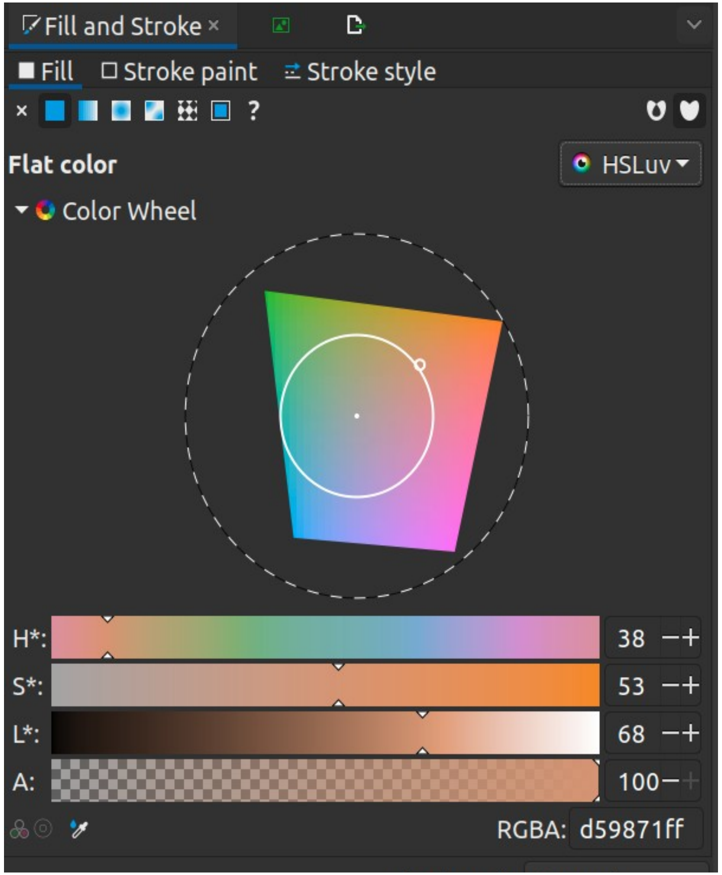




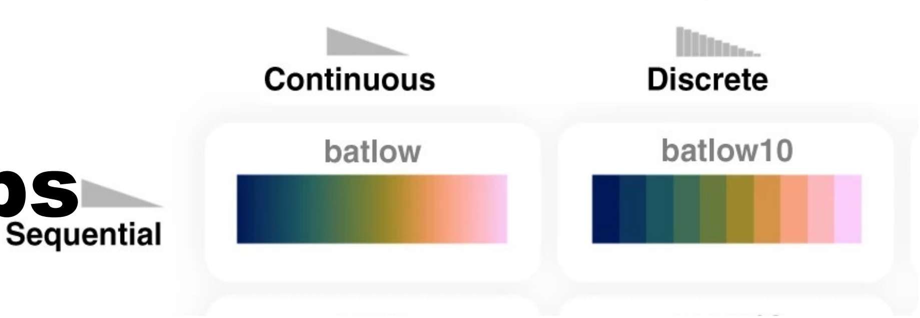
HSL vs H*S*L* (or HSLuv)





Hue
Saturation
Lightness / **L**uminance

Types of colormaps



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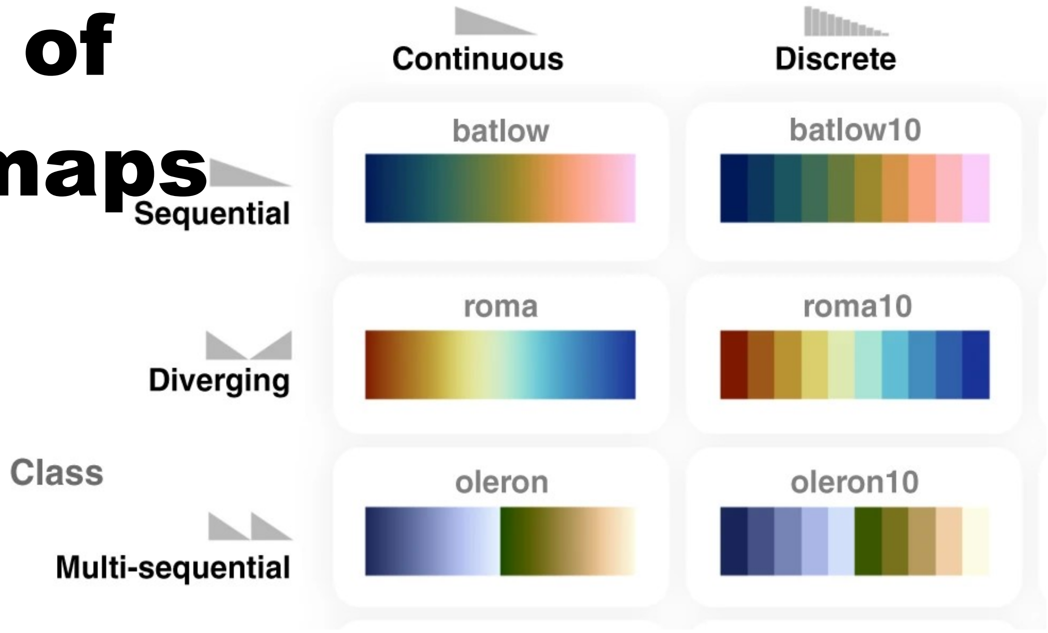

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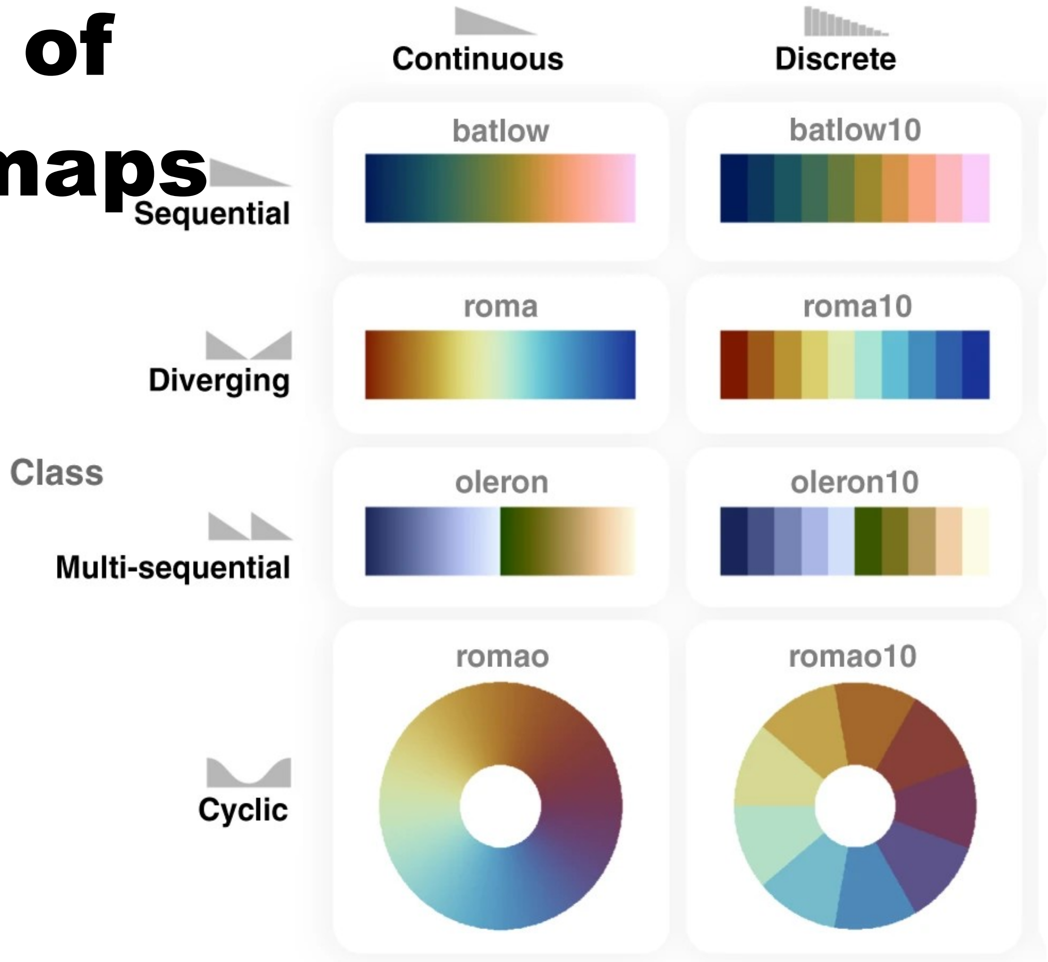

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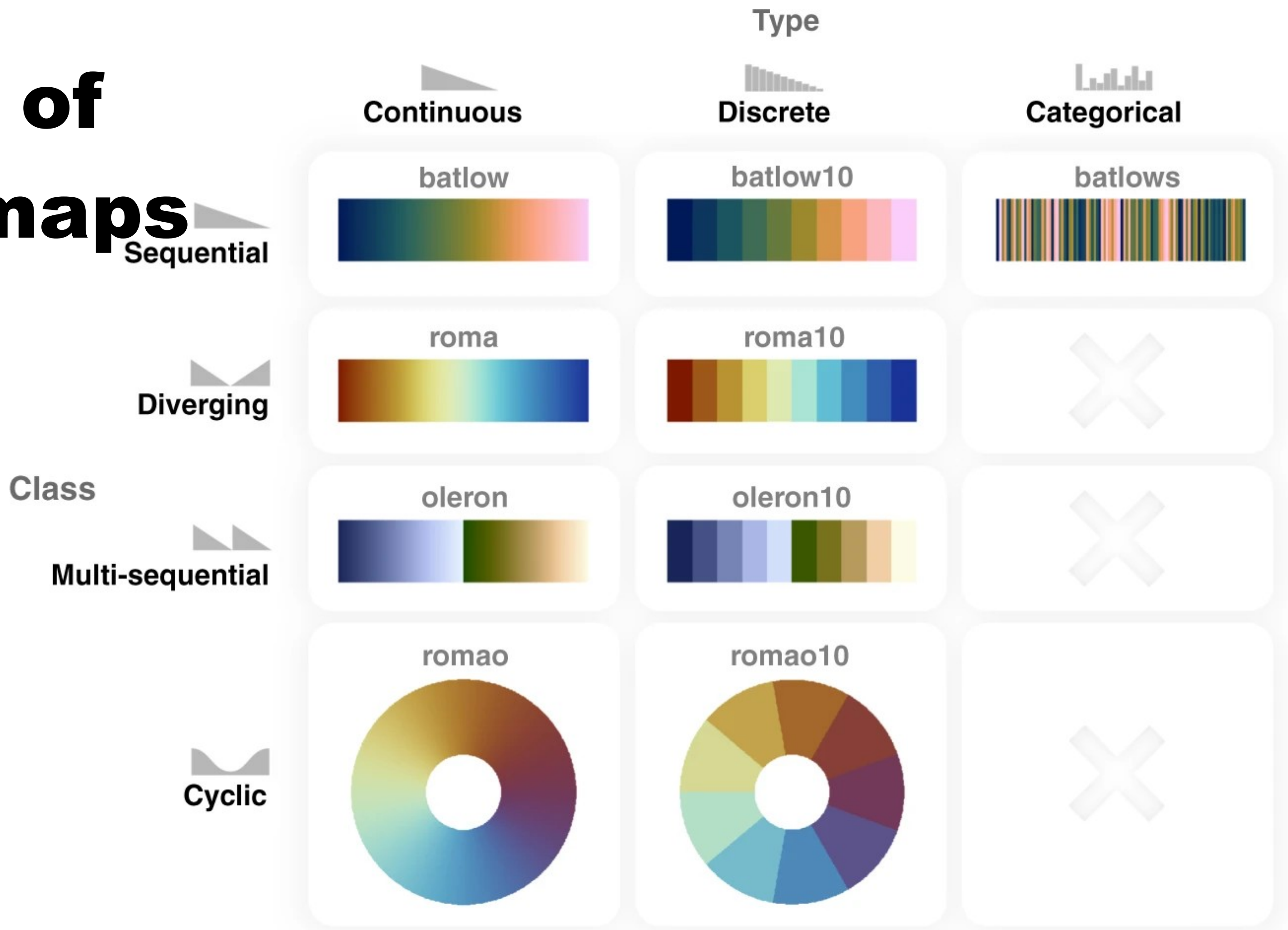

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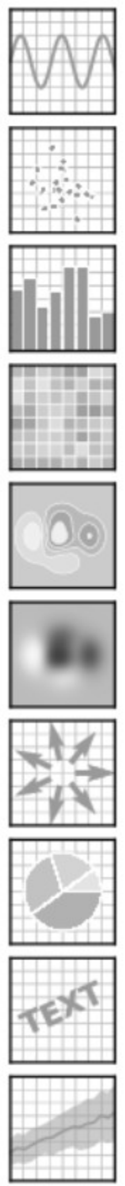
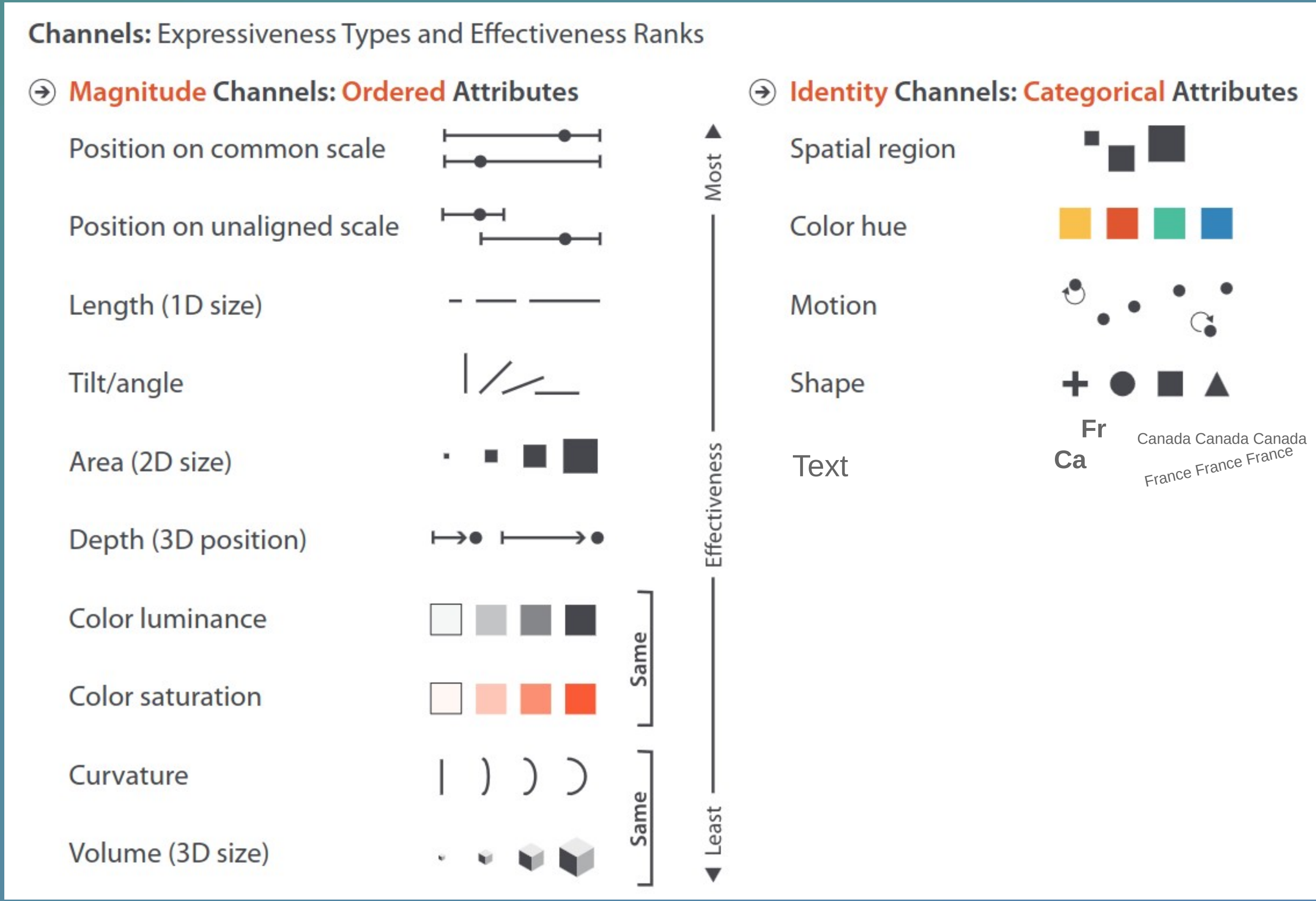
(For future reference)

Choosing effective charts





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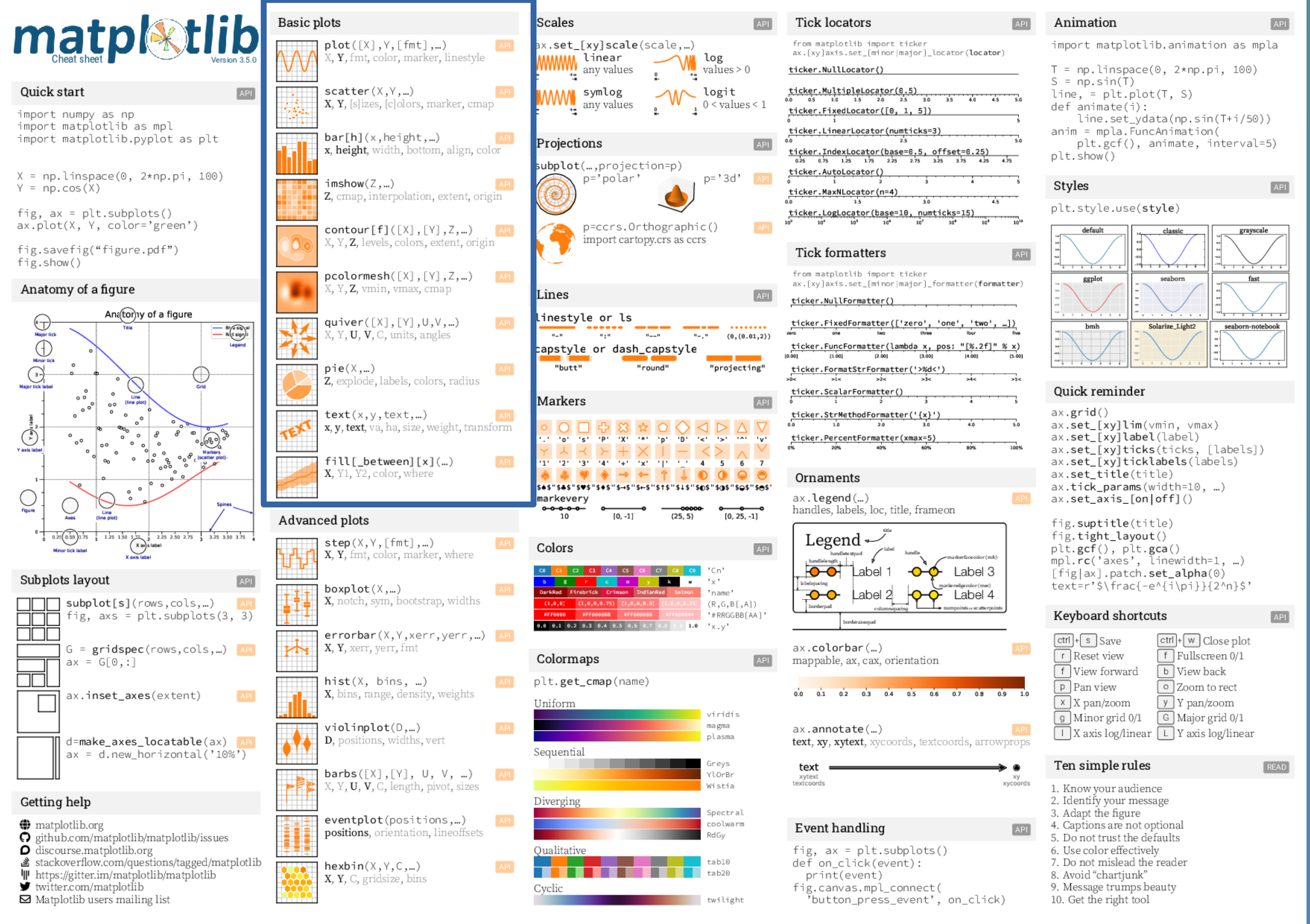

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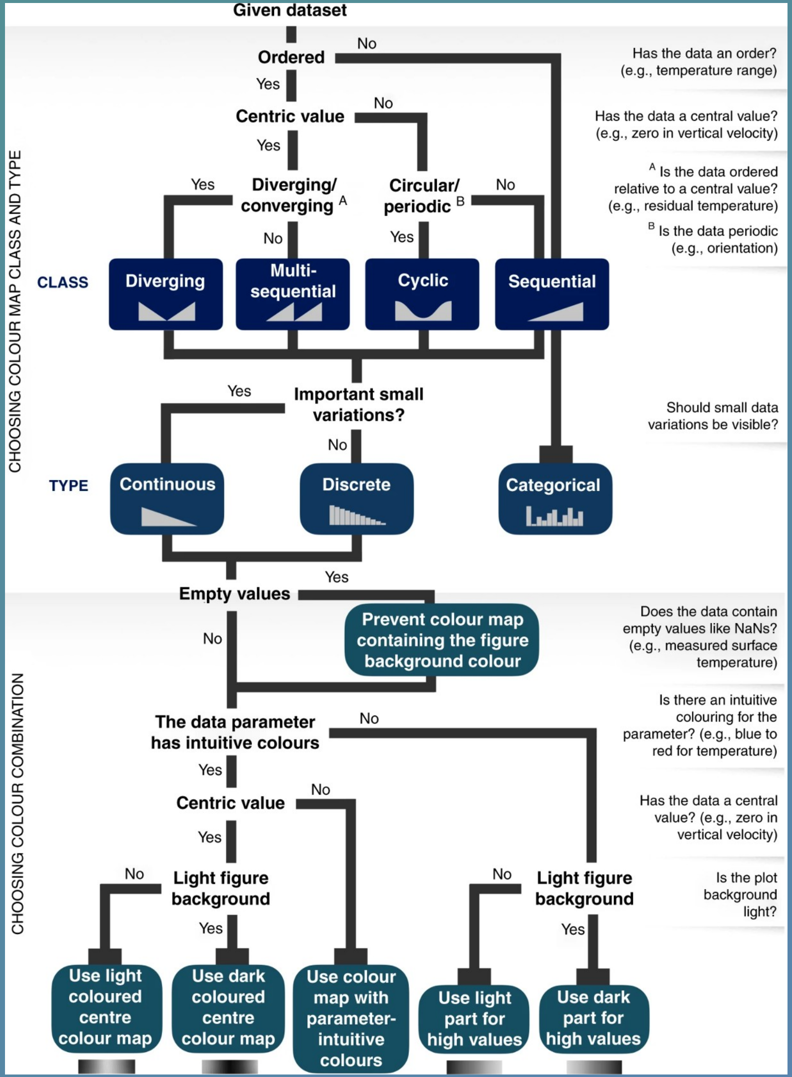

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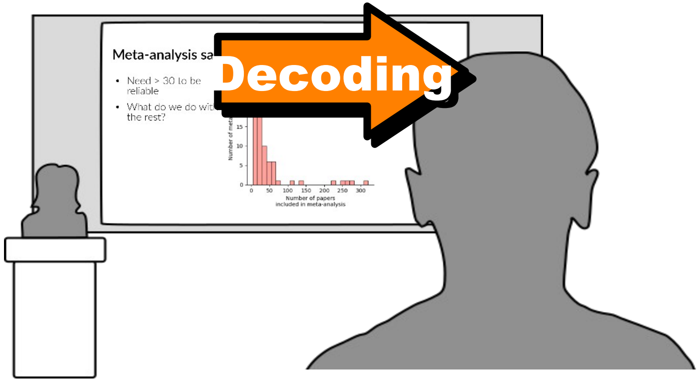
Sequential

For the Sequential plots, the lightness value increases monotonically through the colormaps. This is good. Some of the L^* values in the colormaps span from 0 to 100 (binary and the other grayscale), and others start around $L^* = 20$. Those that have a smaller range of L^* will accordingly have a smaller perceptual range. Note also that the L^* function varies amongst the colormaps: some are approximately linear in L^* and others are more curved.

```
plot_color_gradients('Perceptually Uniform Sequential',
                     ['viridis', 'plasma', 'inferno', 'magma', 'cividis'])
```


```
plot_color_gradients('Sequential',
                     ['Greys', 'Purples', 'Blues', 'Greens', 'Oranges', 'Re
                      'YlOrRd', 'OrRd', 'PuRd', 'RdPu', 'BuPu',
```



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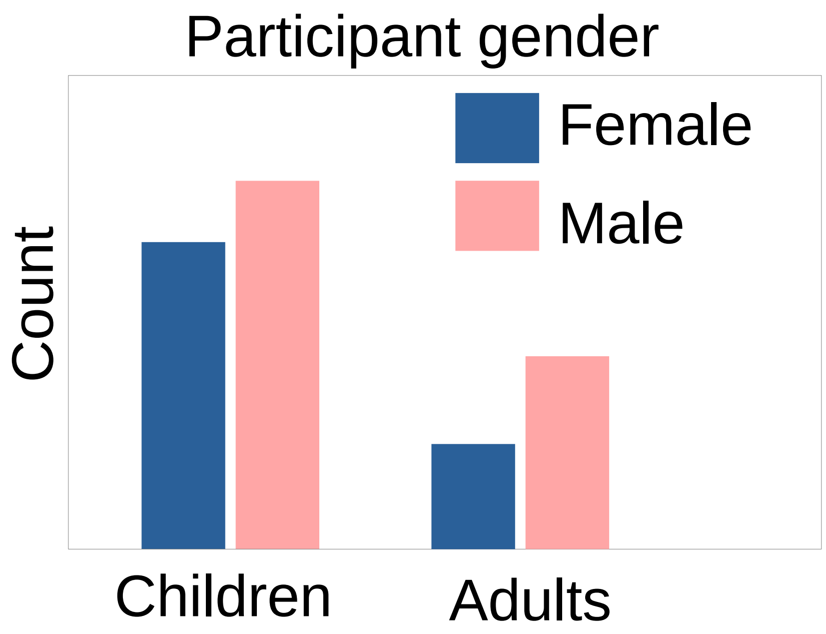
Culture: Color associations



Science: Rainbow colormaps are disliked

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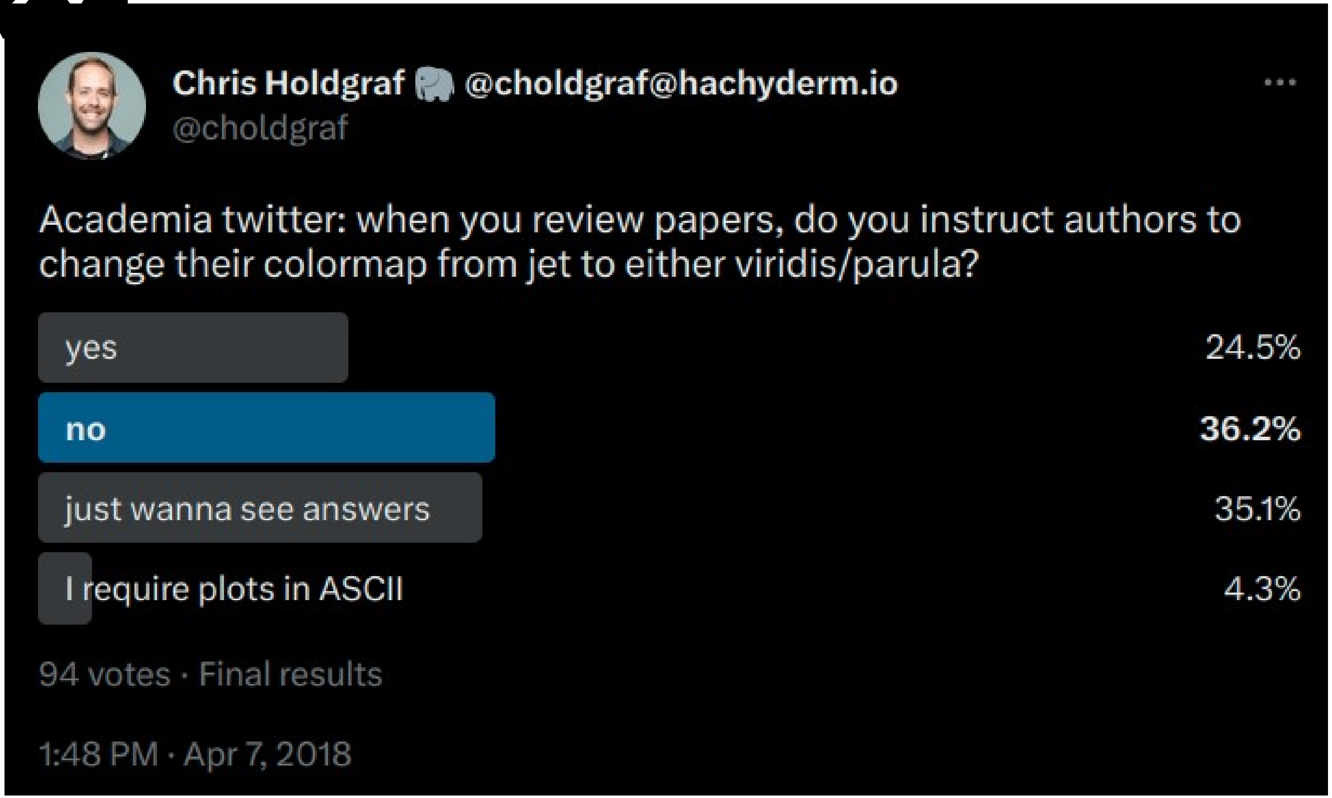
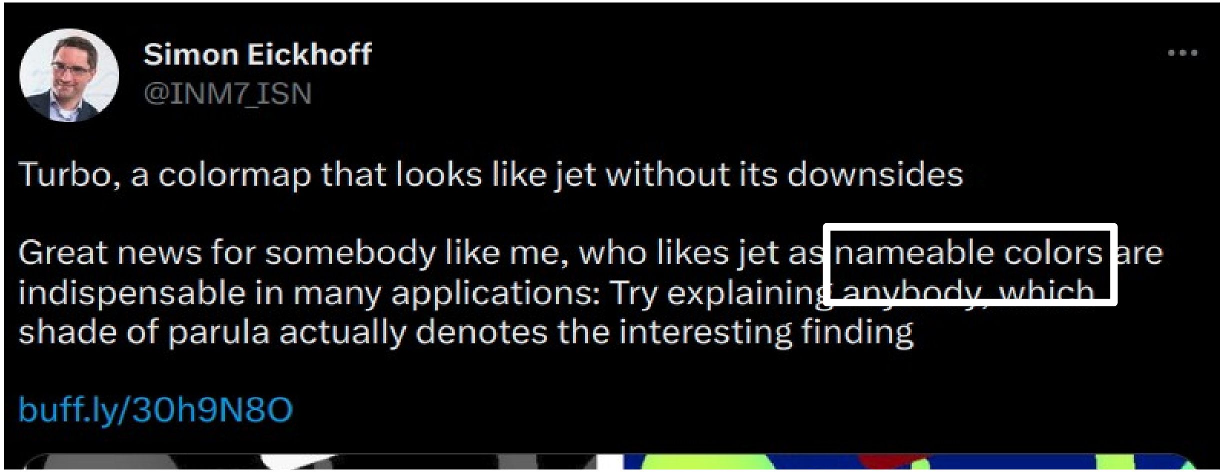
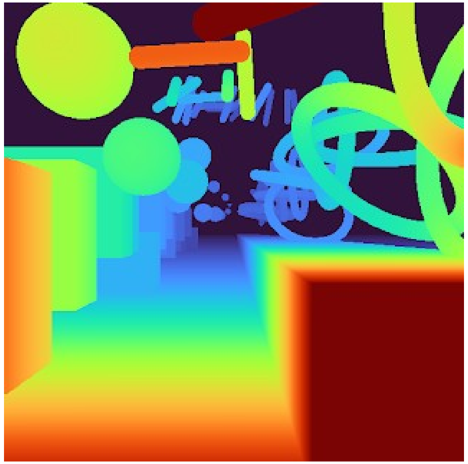


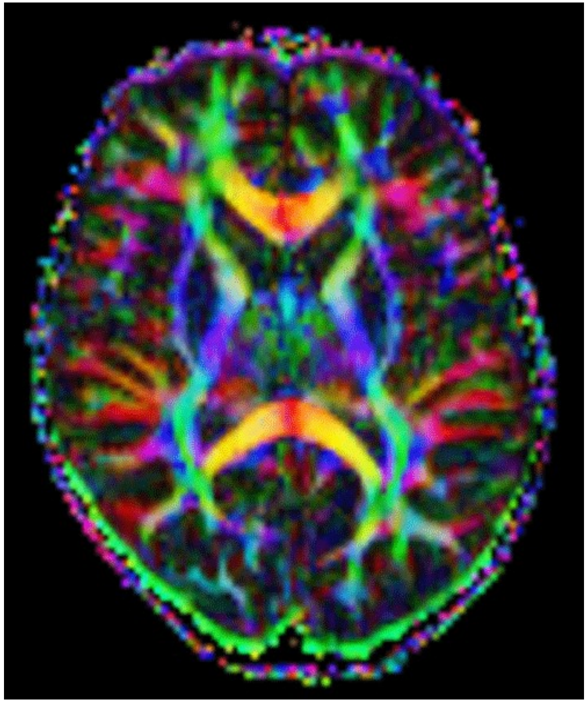
Science: Rainbow colormaps are disliked?

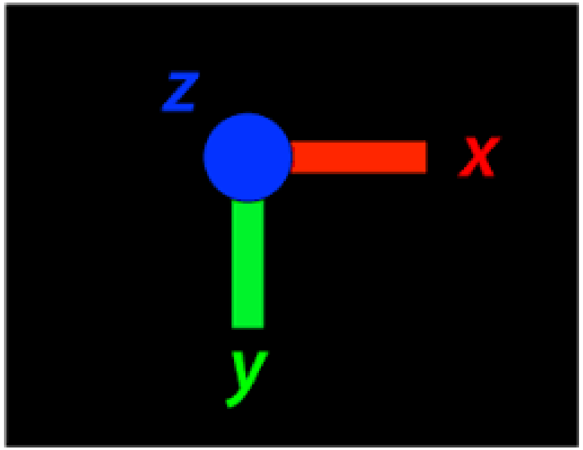




Neuroscience: Colors = 3D direction

Principal Diffusion Direction





https://mriquestions.com/dti-tensor-imaging.html

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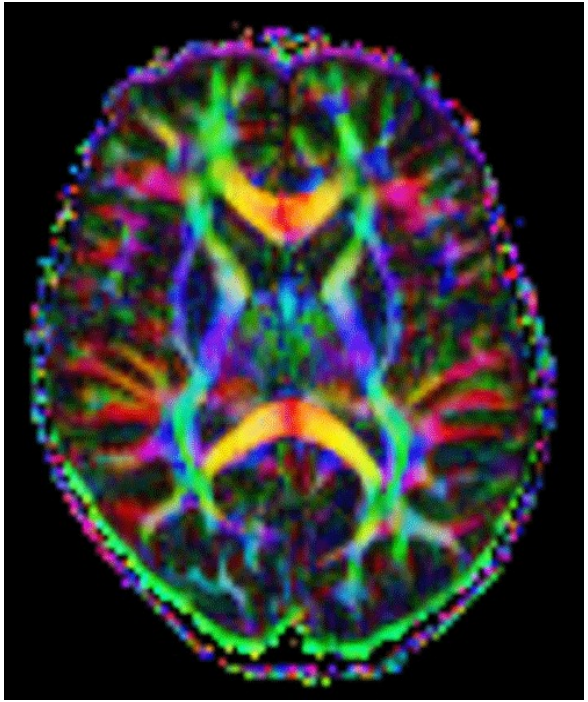
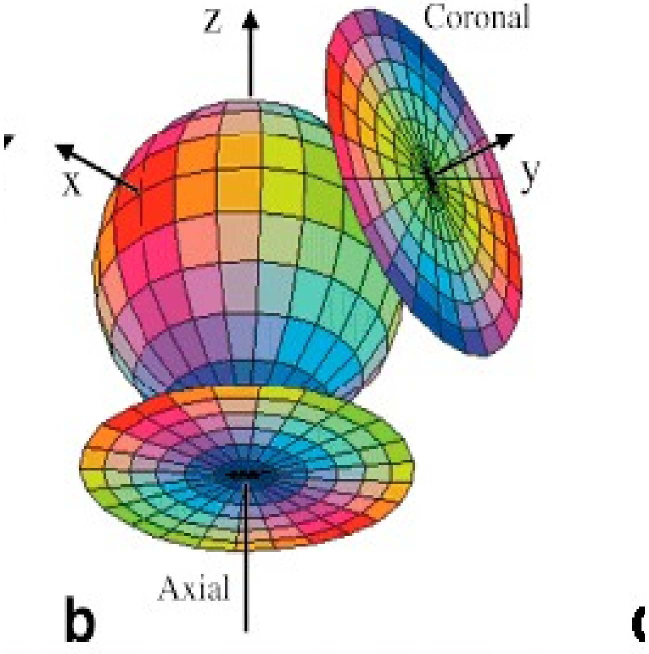




https://mriquestions.com/dti-tensor-imaging.html

https://mipav.cit.nih.gov/pubwiki/index.php/DTI_Color_Display

(For future reference)

Examples of visualization conventions

- Culture
  - Pink is female
  - Time goes left to right
- Science
  - Rainbow colormaps are disliked
- Neuroscience
  - MRI in grayscale
  - DWI colors = directions

Example 1 – Simple data and figure

- Original paper on the ABIDE dataset
  - 964 subjects
    - 396 male
    - 51 female

Lets visualize this

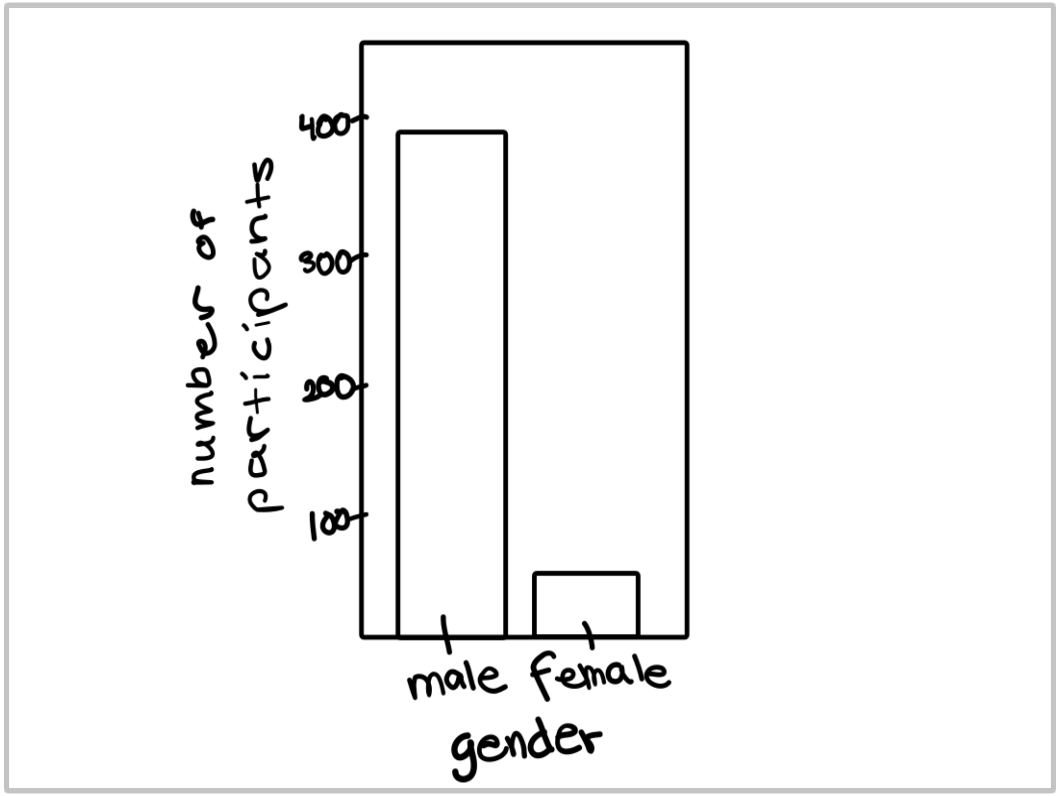
Nielsen, J. A., Zielinski, B. A., Fletcher, P. T., Alexander, A. L., Lange, N., Bigler, E. D., ... & Anderson, J. S. (2013). Multisite functional connectivity MRI classification of autism: ABIDE results. Frontiers in human neuroscience, 7, 599.

What's our message?

- More male than female participants
  - 2 categories, 2 values

How should we communicate this message?

- Clearly show 2 categories and 2 different values



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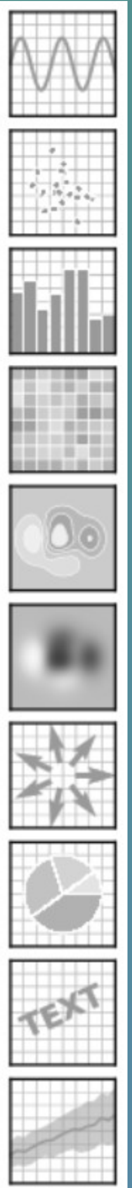

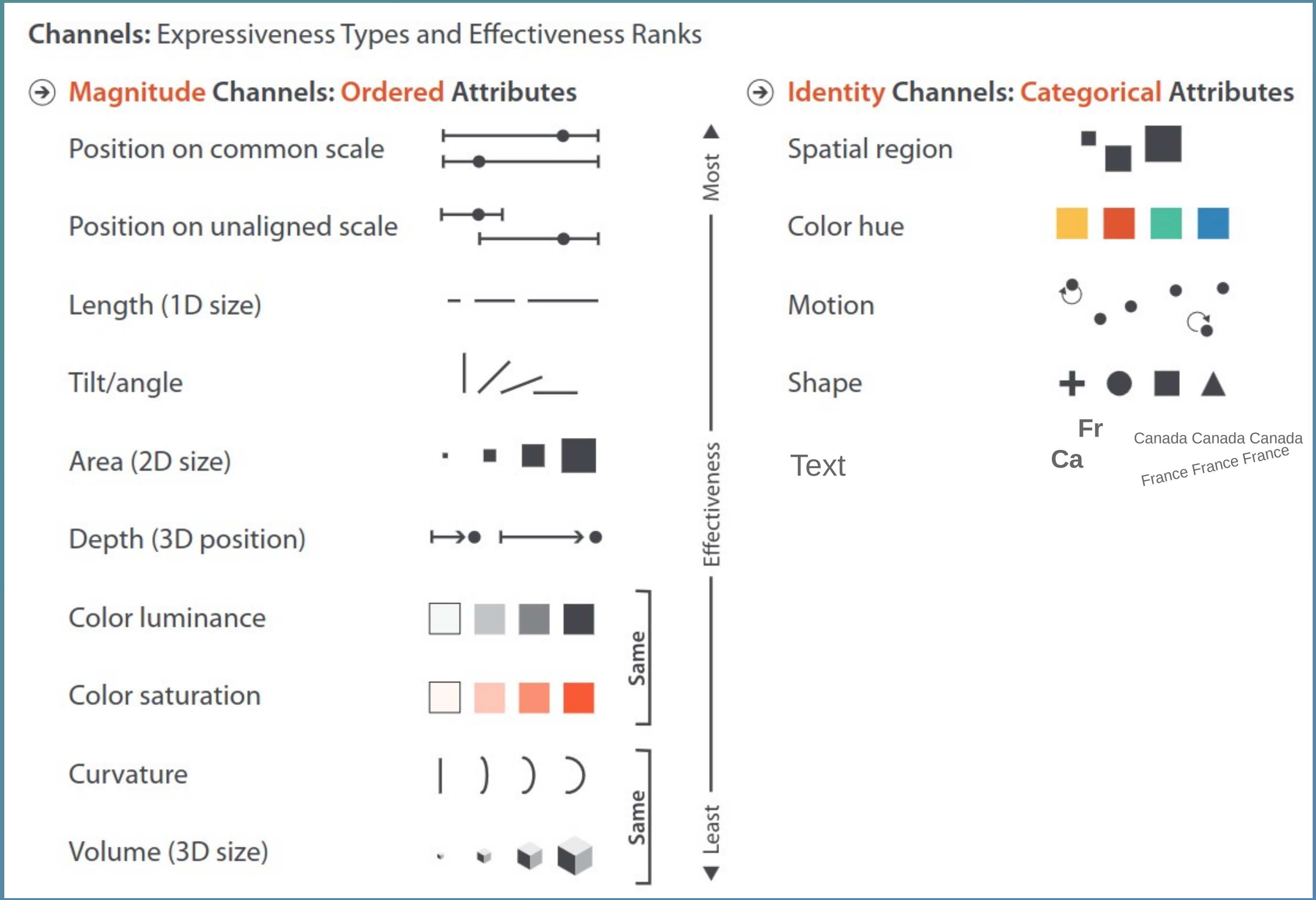

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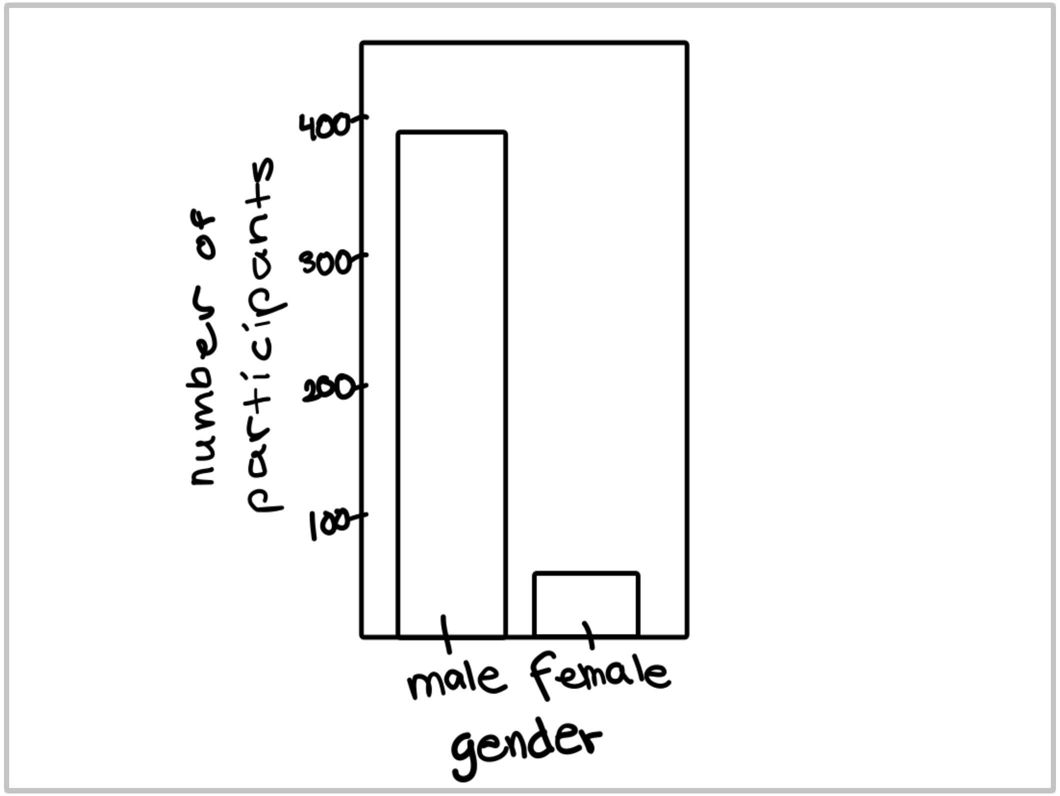

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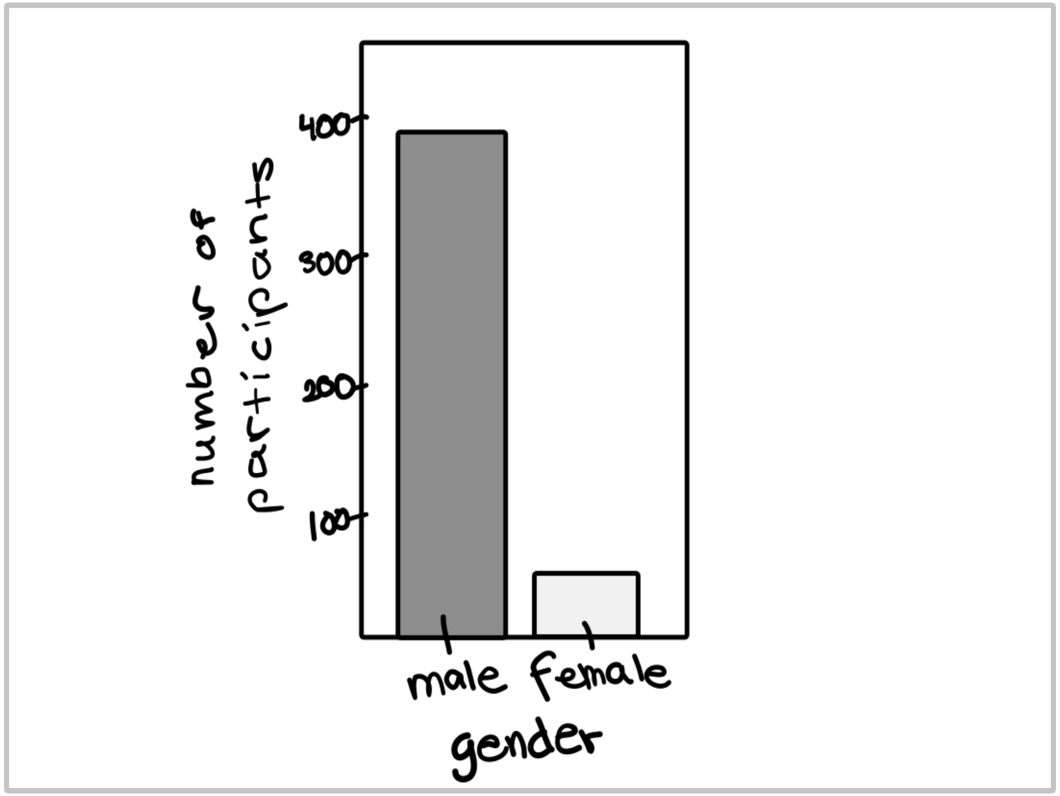
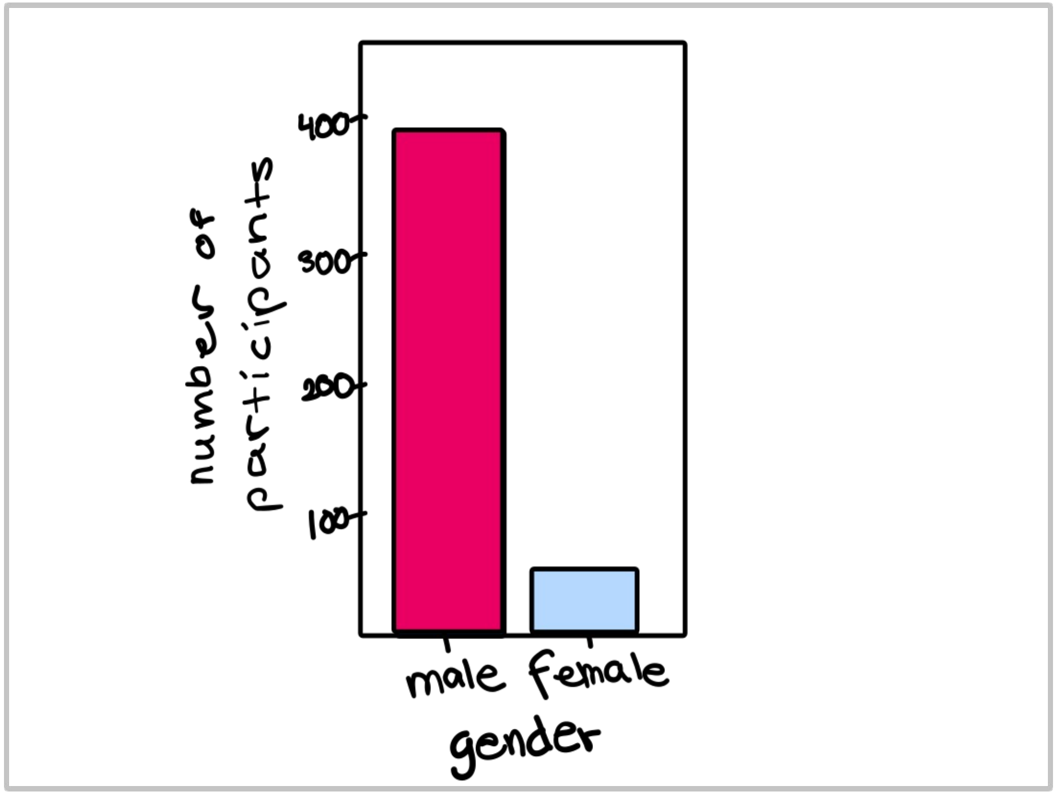


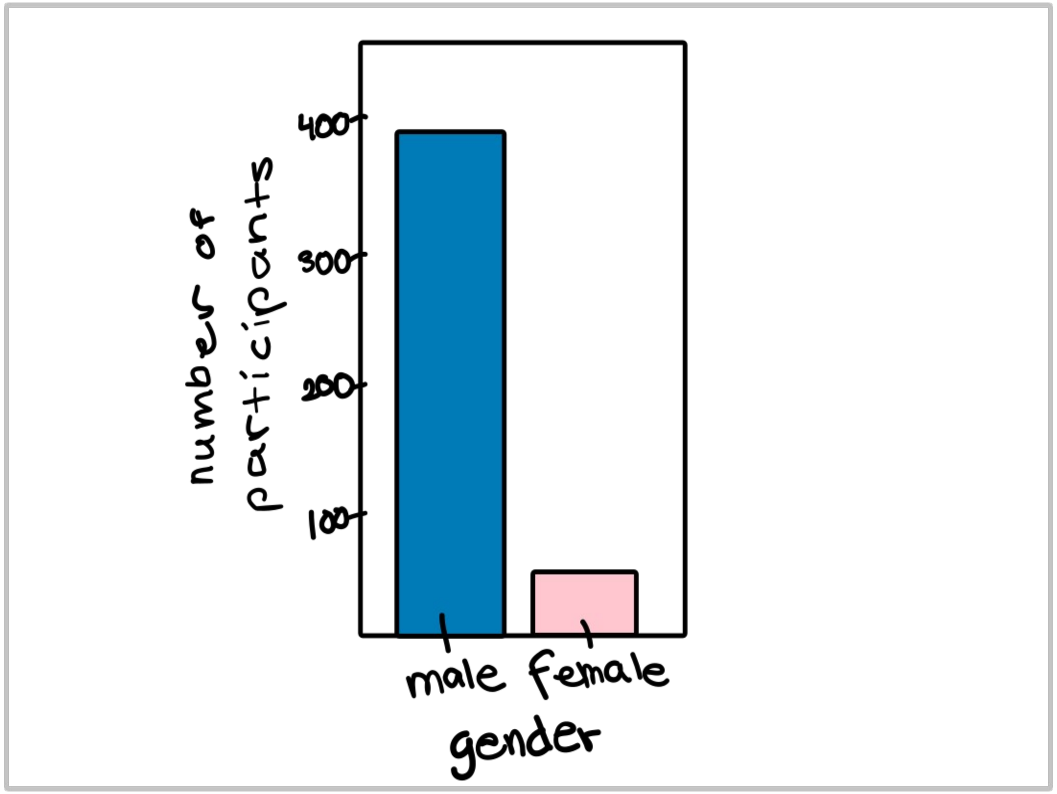
How should we communicate this message?

- Clearly show 2 **categories** and 2 different values



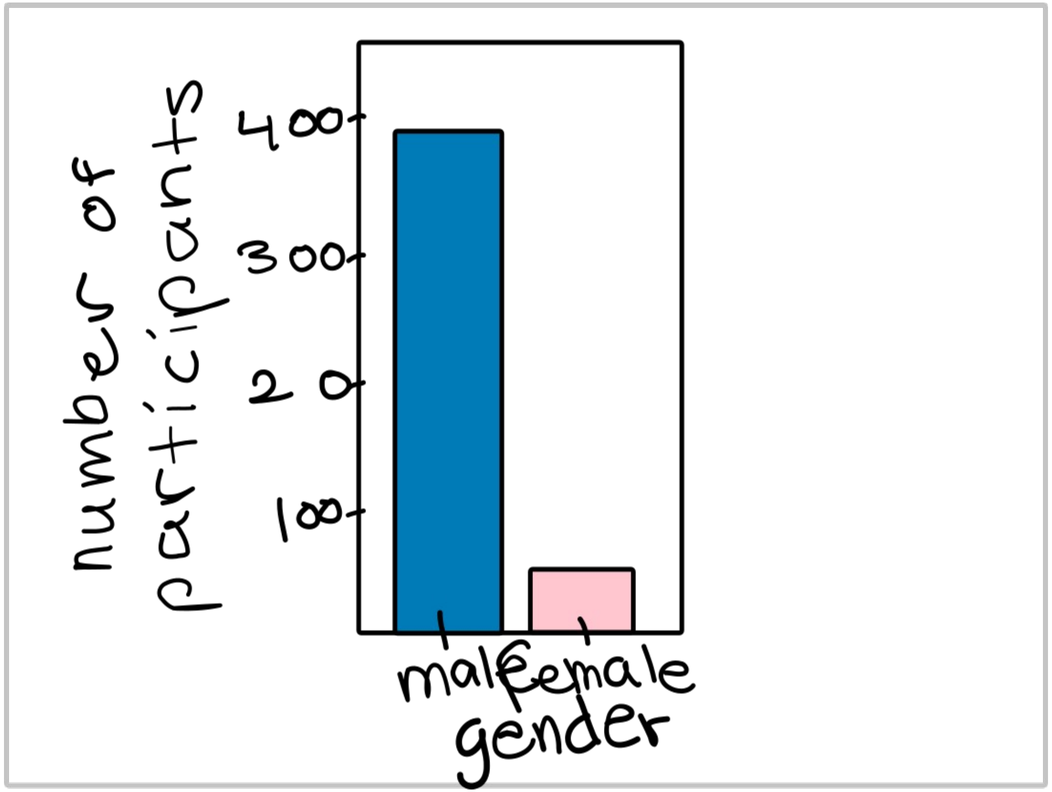
Are there any conventions we should think about?

- Gender & color



Do we need to adapt it to the context?

- Academic presentation



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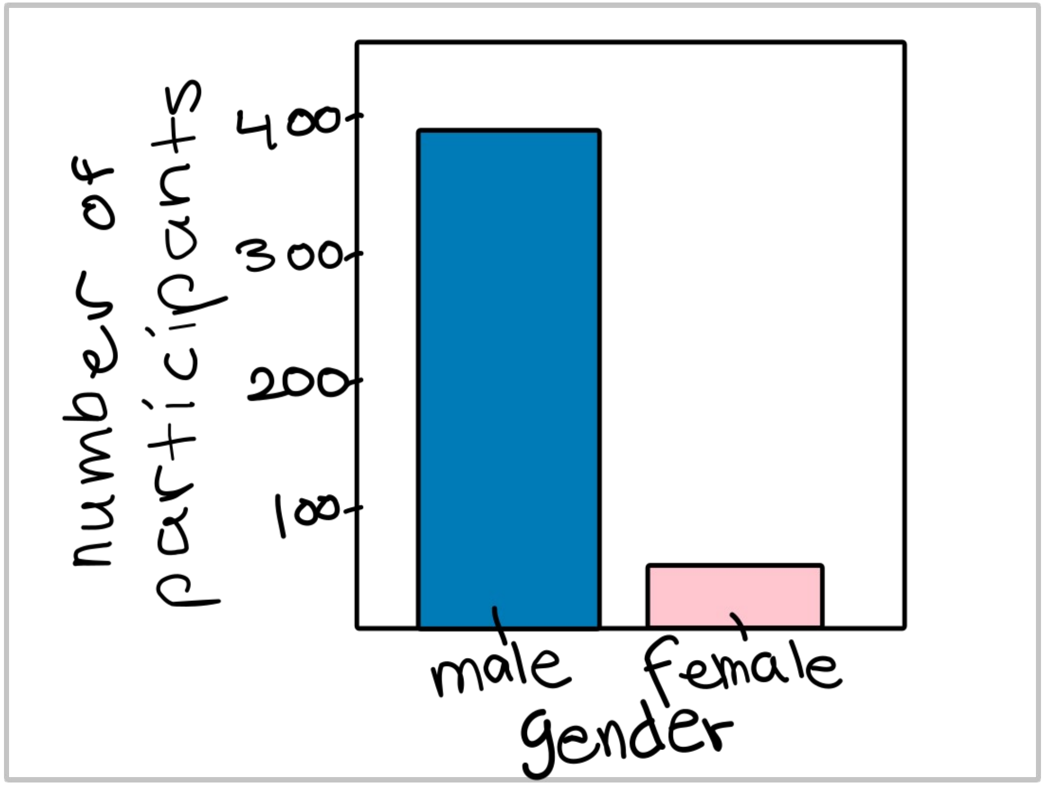

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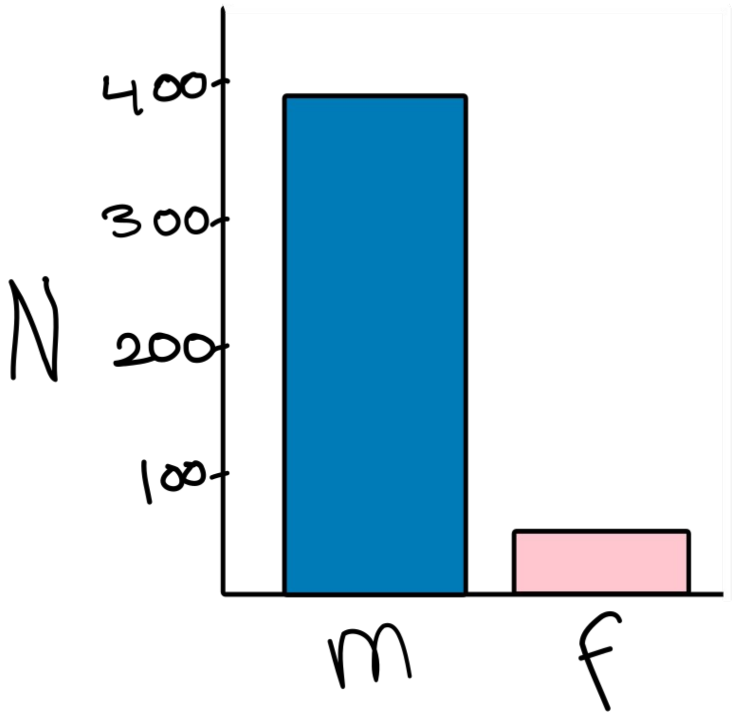
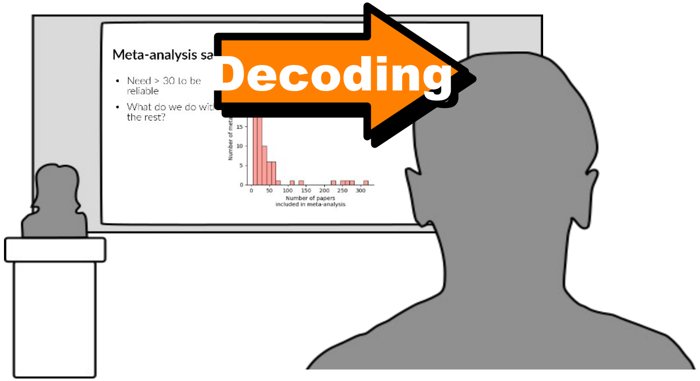




To plan an **effective visualization**, we need to think about

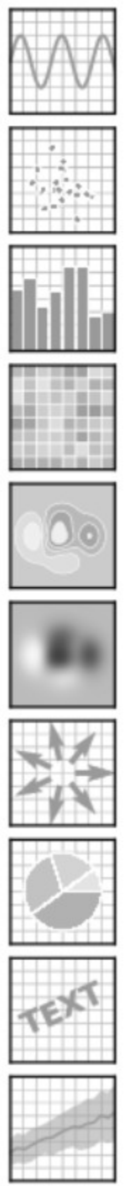
- **Message**
  - **What** we want to communicate
- **Perception**
  - **How best** to communicate it
- **Conventions**
  - **How** it's **usually** communicated

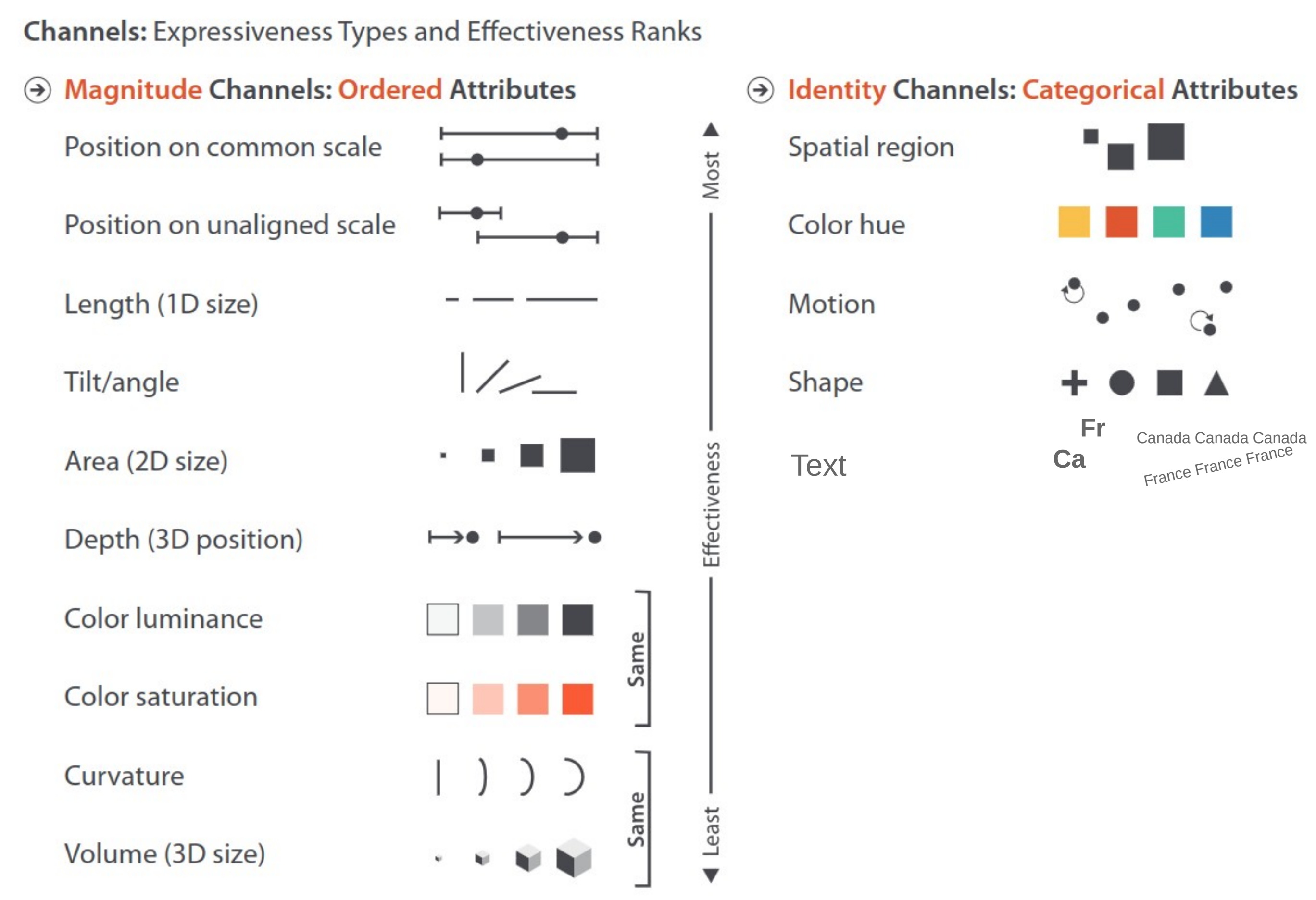
The end of part 1

Reference slides...

(For future reference)

Choosing effective charts



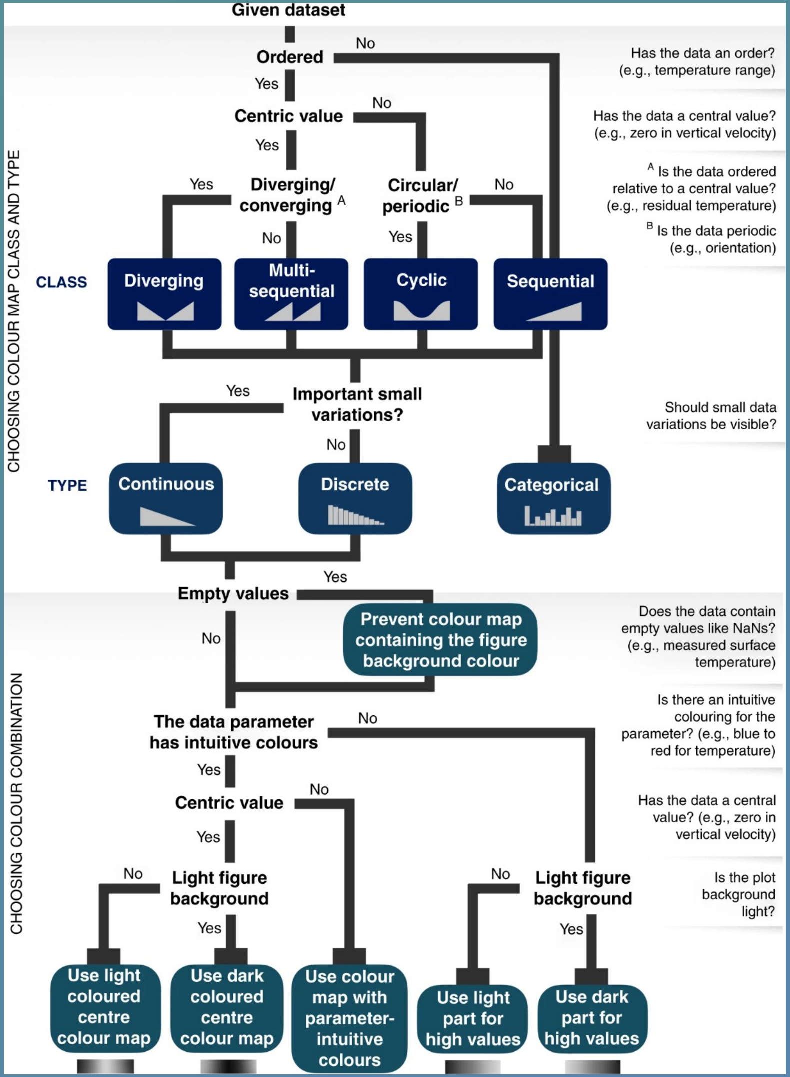


https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png

Munzner, T. (2014). Visualization analysis and design. CRC press.

(For future reference)

Choosing effective colormaps



Crameri, F., Shephard, G. E., & Heron, P. J. (2020). The misuse of colour in science communication. Nature communications, 11(1), 5444.

(For future reference)

Examples of visualization conventions

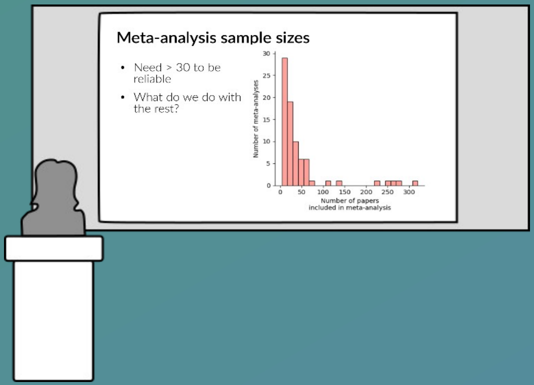
- Culture
  - Pink is female
  - Time goes left to right

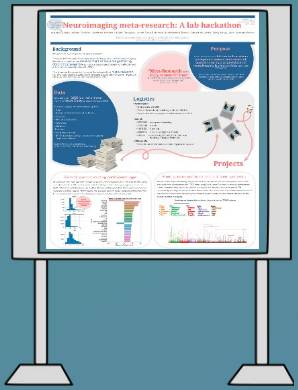
- Science
  - Rainbow colormaps are disliked

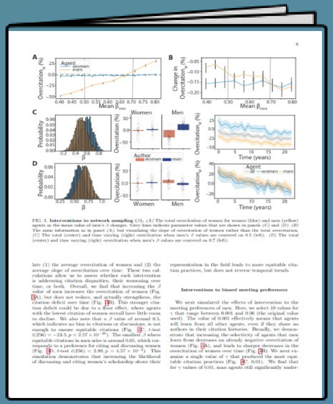
- Neuroscience
  - MRI in grayscale
  - DWI colors = directions

(For future reference)

Contextual adjustments







Less

Time to understand

Proximity to viz

Text explanations

Self-explanatory

Visual complexity

More